Aluminium-Water Combustion for Zero Carbon Marine Shipping: Masters Proposal

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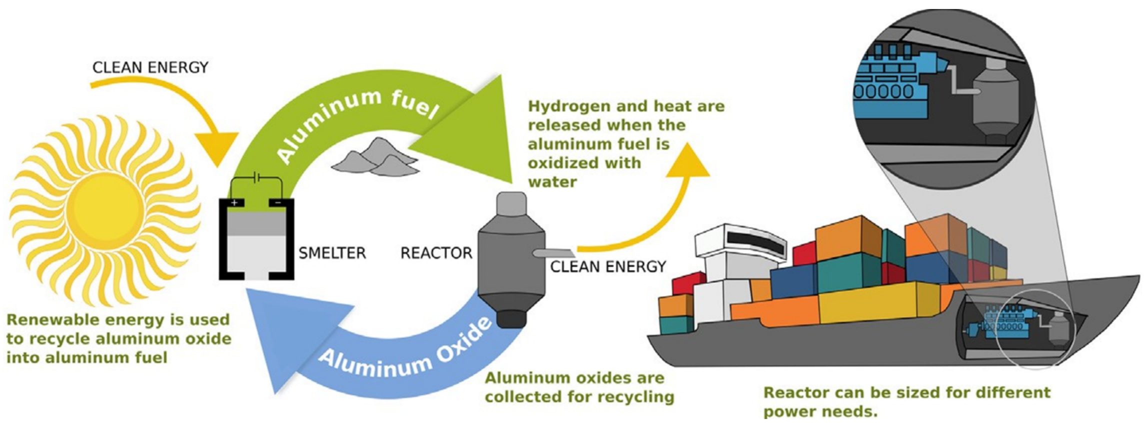

Figure 1: Aluminium powder as maritime shipping fuel (Energy Life-cycle). Adapted from [1]

Abstract

Maritime shipping accounts for 2–3% of global greenhouse gas emissions, with deep-sea routes responsible for approximately 70% of the sector's fuel consumption. Existing electrofuel alternatives exhibit significantly lower volumetric energy densities than heavy fuel oil, and lithium-ion batteries remain impractical at ocean scale. This study investigates aluminium-water combustion via a high-temperature metal-water reactor (HT-MWR) as a zero-carbon marine propulsion technology, addressing four gaps in the published literature.

First, no validated engineering requirements dataset exists for aluminium-water propulsion in any marine vessel class. This study produces the first such matrix through semi-structured interviews with superyacht builders, naval architects, classification societies, and vessel owners. The superyacht segment is identified as the optimal commercialisation beachhead: unlike container shipping—requiring a decade-long IMO approval process—superyacht classification operates under a flexible, risk-based framework. This regulatory contrast, combined with Ultra-High-Net-Worth client tolerance for a green premium, mirrors Tesla's pathway and provides a credible route to deep-sea deployment.

Second, and centrally, prior HT-MWR research has addressed exclusively stationary baseload applications. This study provides the first characterisation of HT-MWR dynamic behaviour under marine propulsion conditions, extending open-loop transient data to closed-loop control system validation. A microcontroller-based controller quantifies dynamic performance across discrete power modes, measuring response lag, rise time, settling time, overshoot, and stability boundaries. The experimental data enables mode granularity optimisation and bounds battery energy and power requirements for future hybrid-electric integration.

Third, the influence of seawater chemistry on HT-MWR performance has not previously been studied. A four-category water treatment design—from distilled water through synthetic and natural seawater—isolates ionic, biological, and particulate effects on hydrogen yield, thermal output, and by-product composition. A tiered approach prioritises distilled water experiments as the core deliverable.

Fourth, a systematic trade-off analysis compares candidate prime movers and thermal integration pathways, using transient data to specify the minimum viable battery buffer. Together, these contributions establish the technical and commercial foundation required for the first prototype aluminium-water marine propulsion system.

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1 Research Background

Overview

Maritime shipping plays a crucial role in global trade, facilitating around 80% of all commerce [21]. This makes it essential for globalization, economic development, and maintaining living standards. However, the sector also contributes significantly to global greenhouse gas emissions, accounting for 2-3% of global emissions and approximately 10% of all transport-related emissions [22, 21].

In response, the International Maritime Organization (IMO) has set ambitious targets to curb emissions from international shipping. By 2030, the IMO aims to reduce annual greenhouse gas emissions by 20-30% compared to 2008 levels, with a more aggressive target of 70-80% reduction by 2040 [23].

1.1 Preliminary Analysis of Electro-Fuels Options

Overview

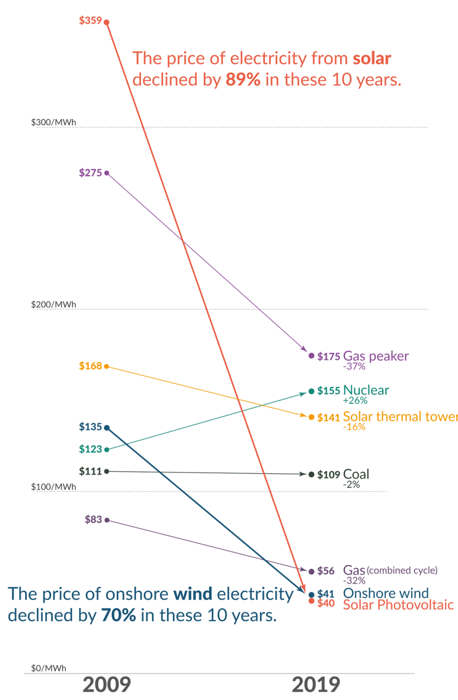


Figure 2: Falling cost of solar electricity vs Fossil Fuels [12]

Recent years have witnessed a dramatic decline in the cost of low-carbon electricity sources, particularly solar and wind power, as illustrated in Figure 2. This cost reduction has significantly enhanced the viability of transitioning to a low-carbon economy, yet substantial challenges persist in the so-called hard-to-abate sectors, among which marine shipping features prominently.

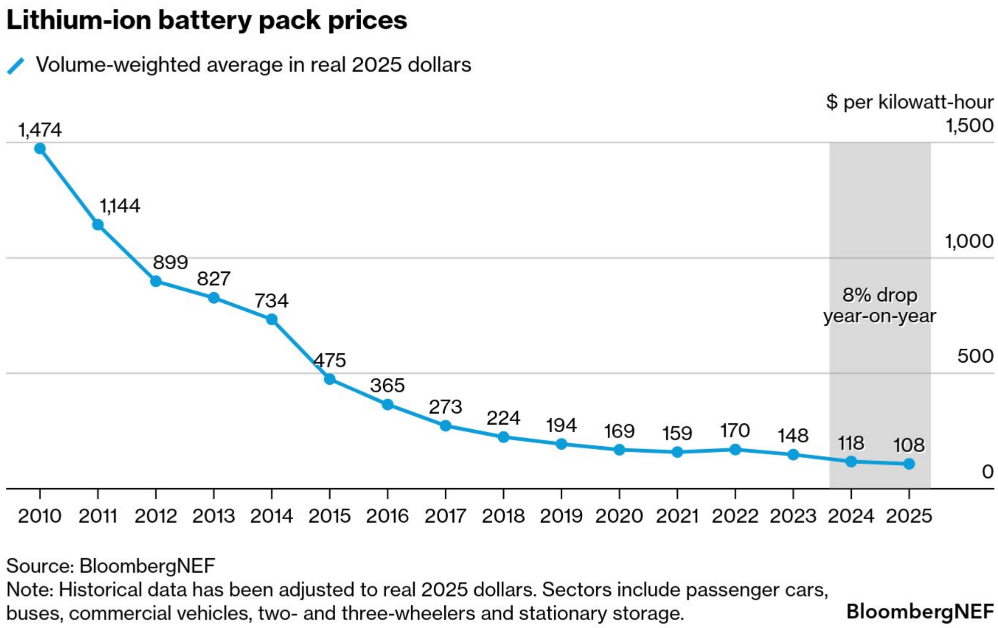


Figure 3: Falling cost of lithium ion battery packs [13]

The predominant method for storing electrical energy remains electrochemical batteries, with lithium-ion technology leading the charge. These batteries have increasingly displaced fossil fuels in passenger vehicles and, more recently, have begun penetrating short and medium-distance trucking applications [24]. This transformation has been driven largely by the precipitous decline in lithium battery prices, as demonstrated in Figure 3. The technology's reach has even extended to inland shipping routes, where standardized and swappable battery packs are being deployed in China [25]. Major battery manufacturers such as CATL are making substantial investments in these markets, signaling confidence in the technology's continued expansion [26].

However, despite remarkable advances in both price and energy density, lithium-ion batteries remain largely impractical for deep-sea shipping, which accounts for approximately 70% of marine shipping emissions [27]. The fundamental constraint lies in the current energy density limitations of lithium-ion technology, which ranges from 0.1 to 0.5 kWh/kg [2, 3]—substantially lower than gasoline, which can exceed 10 kWh/kg [6]. This considerable disparity in energy density presents a formidable barrier to the electrification of long-distance maritime transport.

In response to the energy density limitations of lithium-ion batteries, researchers and policymakers have increasingly explored the use of abundant and inexpensive solar electricity to produce alternative fuels, primarily through electrolysis. This approach has given rise to a diverse landscape of electro-fuel options, which can be systematically organized into four broad classifications:

- **Simple electro-fuels** - hydrogen and ammonia, which represent the most straightforward products of electrolytic processes

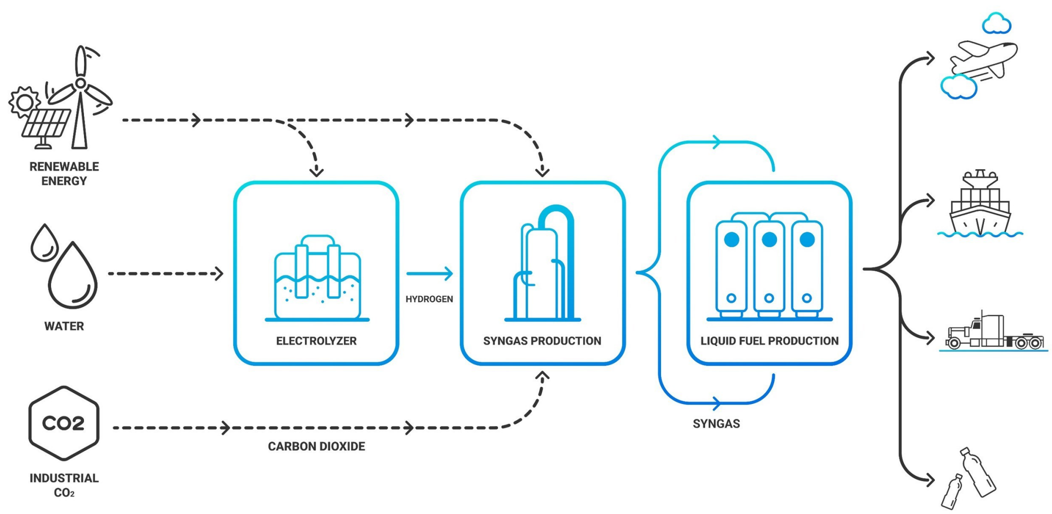


Figure 4: High Level Process Diagram for synthetic hydrocarbon production from renewable electricity [14]

- **Synthetic hydrocarbons** - primarily methane and methanol, whose production pathways are illustrated in Figure 4, which depicts the high-level process for synthesizing these fuels from low-carbon electricity

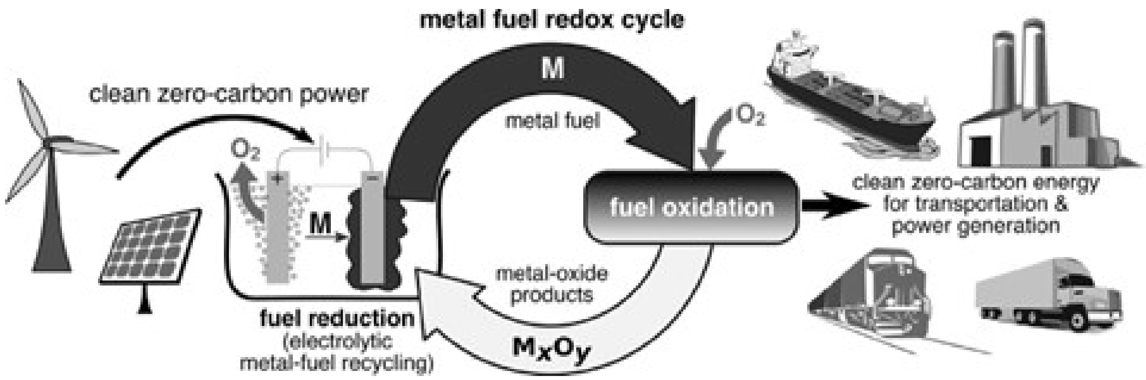


Figure 5: Metal based electro-fuels as secondary energy carriers [15]

- **Metal-based electro-fuels** - including iron, aluminium, magnesium, silicon, and boron. Figure 5 demonstrates the high-level process by which these metal-based electro-fuels can function as recyclable secondary energy carriers, offering a distinctive closed-loop energy storage paradigm
- **Metal hydrides (hydrogen carriers)** - such as MgH_2, AlH_3, $LiBH_4$, and $NaAlH_4$, providing an alternative means of storing and transporting hydrogen in a more dense and stable form than its gaseous state

In the following subsections, a preliminary analysis is conducted to compare the mass energy and volumetric energy densities of these various electro-fuel options in order to understand their viability for marine shipping applications.

1.1.1 High Level Energy Density Analysis

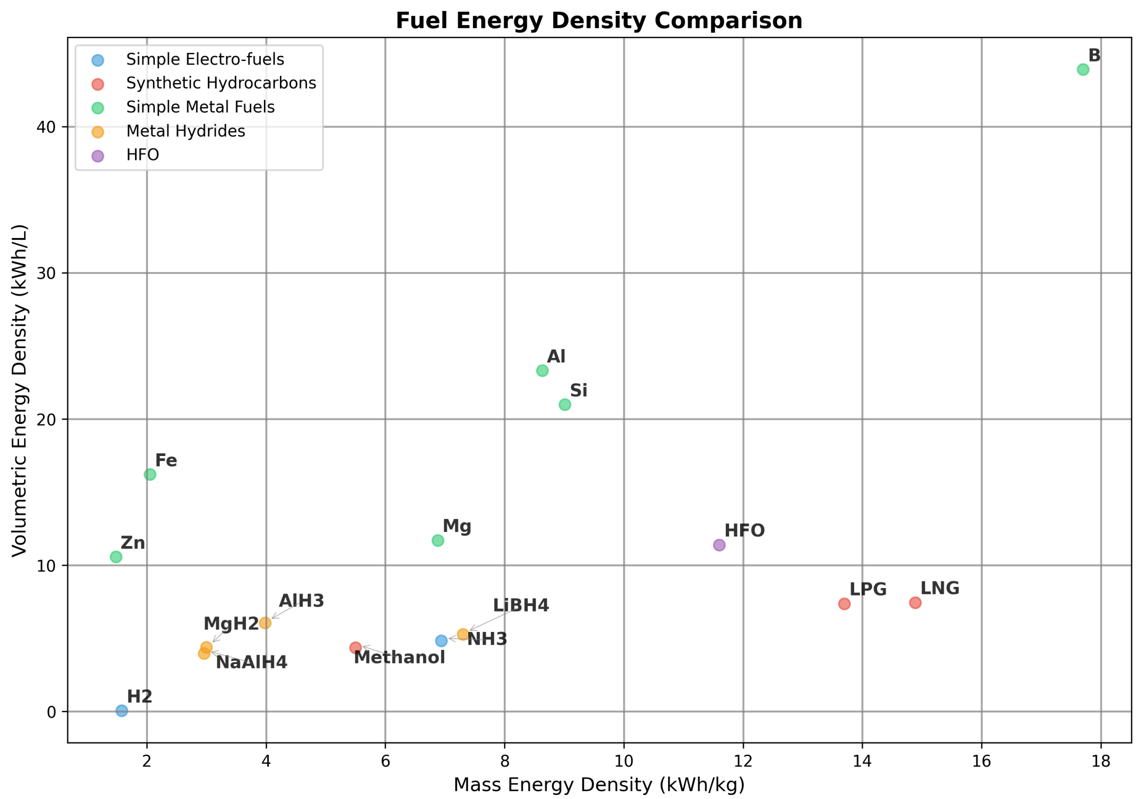


Figure 6: Simplified Energy Density Analysis. Data Analysis can be found in the Appendix. Adapted from [16]

Note: 1) Pressure Cylinder H2 is 25°Celsius 500 bar
2) Liquid NH3 is -33°Celsius 1 bar. Energy needed for cracking for NH3 to H2 is ignored in this table and all subsequent calculations.

Figure 6 presents the fuel energy density calculations for the various electro-fuel options currently under investigation for marine applications. Heavy Fuel Oil (HFO) serves as the reference fuel in these comparisons, as it remains the dominant fuel source for global marine shipping due to its low cost and widespread availability.

Hydrogen, which is being explored as a potential solution for several hard-to-decarbonise sectors including shipping and aviation, requires pressurisation at approximately 500 bar. This compression requirement drastically diminishes the fuel's energy density, presenting significant challenges for practical implementation. Even methanol, which is considered a relatively straightforward drop-in replacement for marine shipping and has garnered substantial support from major shipping companies such as Maersk [28], exhibits significantly lower mass and volumetric energy densities compared to HFO.

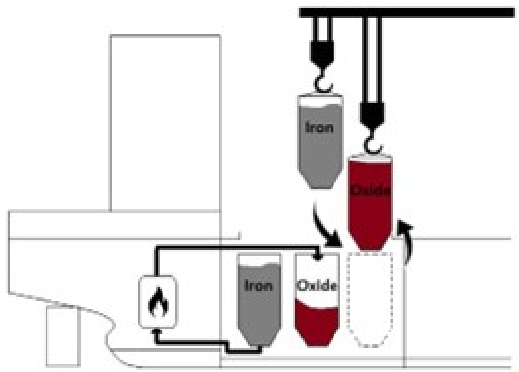


Figure 7: Proposed bunkering procedure for metal fuels (iron powder in this specific case) [17]

As illustrated in Figure 6, metal fuels demonstrate considerable promise in terms of both mass and volumetric energy densities, despite the relatively limited research effort devoted to their development. All metal fuels examined display higher volumetric energy densities than HFO, with boron demonstrating superior performance across all available electro-fuel options and surpassing even HFO itself. However, metal fuels present numerous complications that must be addressed for practical deployment. To make metal fuels economically viable, the metal oxide by-products generated during combustion cannot simply be discharged into the ocean. Instead, these by-products must be retained onboard the vessel and offloaded at ports for subsequent recycling using renewable energy sources, as shown in Fig 5. The bunkering procedure for a metal fuels system is depicted in Figure 7. It is crucial to recognise that energy density calculations which fail to account for the mass and volume of the metal oxide by-products can be highly misleading and do not reflect the true practical performance of these fuel systems.

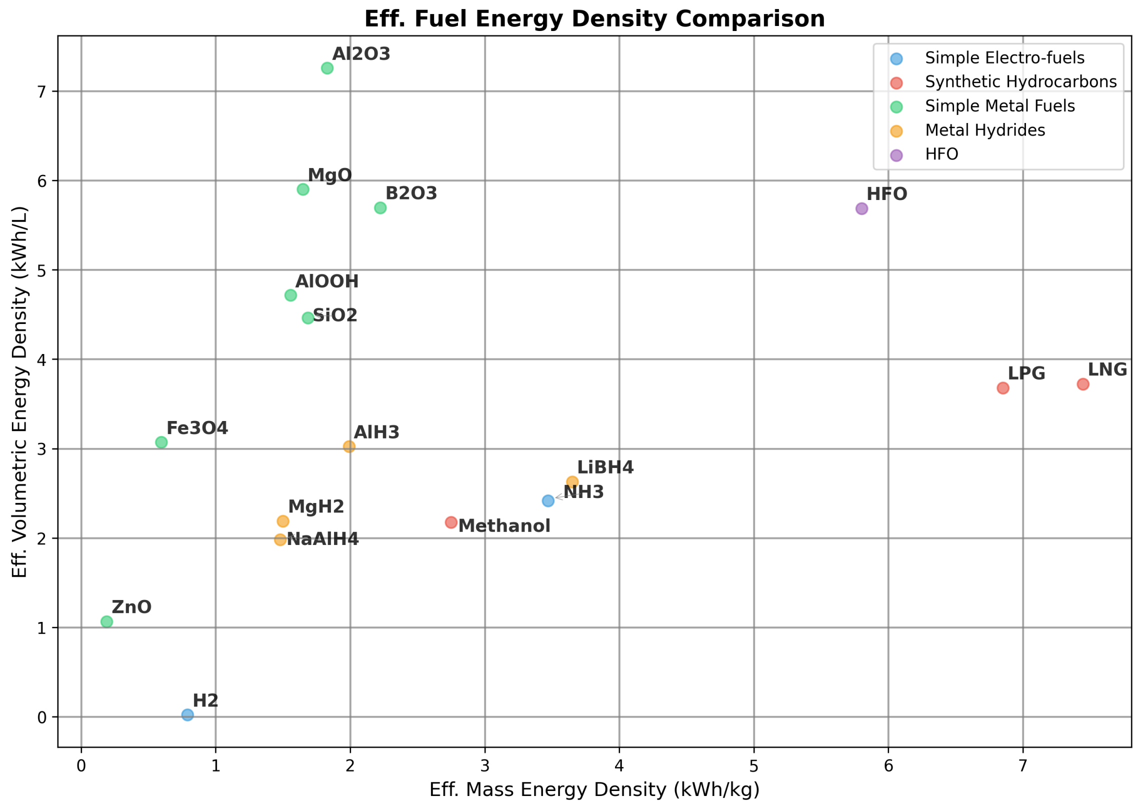


Figure 8: Effective Energy Density Analysis. Data Analysis can be found in the Appendix. Adapted from [16]

Note: 1) Pressure Cylinder H2 is 25°Celsius 500 bar
2) Liquid NH3 is -33°Celsius 1 bar. Energy needed for cracking for NH3 to H2 is ignored in this table and all subsequent calculations.
3) The engine efficiency for hydrogen and hydrocarbon combustion is assumed to be around 50%
4) The engine efficiency for the metal fuels is assumed to be 40%. This number is theortical as no such system currently exists for marine shipping. More information about the metal combustion will be discussed in the following sections.
5) For the metal hydrides, the energy needed to free the stored hydrogen is not considered in this calculation.

1.1.2 Effective Energy Density Analysis

To gain a more comprehensive understanding of the practicality of the different electro-fuel options, an analysis of effective energy densities must be conducted. Effective energy density is defined as the energy density after accounting for engine efficiency, but in the case of metal fuels, it additionally incorporates the weight and volume associated with carrying the metal oxide by-products post-combustion. Figure 8 presents the results from the effective energy density analysis for the various electro-fuel options available. For metal fuels, the effective energy density represents the worst-case scenario of continuously carrying the by-products post-combustion throughout the voyage. In practice, the effective energy density for metal fuels will likely be higher than these calculations suggest, since the by-products can be partially offloaded at ports for collection and subsequent processing at metal reduction facilities.

It should be noted that this is a highly simplified analysis, as the mass and volume of the engines and metal reactors have been deliberately excluded from consideration. Furthermore, there is currently no real-world commercially available marine engine that utilises metal fuels, and therefore an efficiency of approximately 40% has been assumed for these calculations. The complete methodology and detailed calculations for the effective energy density analysis can be found in the Appendix.

As demonstrated in Figure 8, metal fuels remain competitive with the energy density of HFO even after accounting for the metal oxide by-products, particularly with respect to volumetric energy density. Although the mass energy density of the most promising metal fuel candidates, such as aluminium and silicon, remains lower than that of methanol, it is still substantially higher than commercially available electrochemical batteries. The superior volumetric energy densities are of particular significance, given that in practice the metal fuel system will necessitate two sets of storage tanks—one for storing the fuel and another for storing the by-products. This characteristic is also likely to prove advantageous for marine applications where space is at a premium, such as yachts, cruise ships, and container shipping. In later sections, the viability of using aluminium as a marine fuel is discussed in detail, given its promising performance following the effective energy density calculations.

1.2 Aluminium-Water Combustion

Using the energy density analysis from the previous sub-section, aluminium metal emerges as the most promising candidate for use in marine applications. Although it cannot match HFO and other fossil fuels in terms of mass energy density, it demonstrates competitive performance with respect to volumetric energy density. The higher volumetric energy density of aluminium fuel is particularly advantageous for marine applications where space is at a premium, such as yachts, cruise ships, and container ships. Additionally, in practice, the vessel will utilise two storage tanks—one for storing the aluminium powder fuel and another for storing the aluminium oxide by-product, which can be collected later at ports or refuelling stations for recycling.

Aluminium offers several other significant advantages that make it an attractive fuel candidate. It is the third most abundant metal in the Earth's crust, ensuring long-term availability and resource security [1, 29]. Furthermore, aluminium production is already a massive global industry, as the metal's properties make it exceptionally useful across numerous applications. There exists a well-established electrochemical process for producing aluminium from aluminium oxide, known as the Hall-Héroult process. This industrial process can be rendered entirely zero-carbon if the electricity is sourced from renewable energy and the electrolysers make use of inert anodes, thereby eliminating carbon emissions from the production cycle [30].

$$2\,\mathrm{Al} + 3\,\mathrm{H_2O} \longrightarrow \mathrm{Al_2O_3} + 3\,\mathrm{H_2} + 4.21\,\mathrm{kWh/kg{-}Al} \tag{1}$$

$$3\,\mathrm{H_2} + \frac{3}{2}\,\mathrm{O_2} \longrightarrow 3\,\mathrm{H_2O} + 4.40\,\mathrm{kWh/kg{-}Al} \tag{2}$$

Equation 1 demonstrates how aluminium reacts with water to produce heat and hydrogen gas. Equation 2 subsequently shows how this hydrogen gas can then be combusted to produce water and additional heat. Together, these equations reveal that to utilise the full chemical energy potential of the aluminium-water system, both the heat generated from the metal-water reaction and the heat produced from the hydrogen combustion must be harnessed. This metal-water reaction, also known as wet cycle combustion, has already found application in several specialized domains, including ALICE rockets developed for NASA [31] and underwater propulsion applications employed by the American and Russian militaries [32].

1.3 Reactor Design

Overview

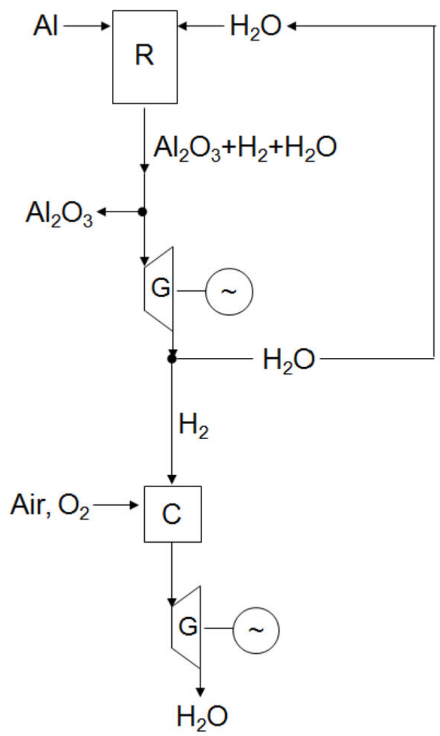


Figure 9: Simplified Overview of Metal-Water Propulsion System[4]
Note: R = metal-water reactor, G = steam generator, C = hydrogen combustion chamber

The metal-water propulsion system, as illustrated in Fig 9, demonstrates a novel approach to maritime propulsion. In this system, seawater from the ocean is mixed with aluminium powder from the fuel tank within a specialized metal-water reactor. The resulting reaction generates three key outputs: hydrogen, thermal heat, and aluminium oxide by-product, with the latter being separated into a dedicated storage tank. The system harnesses energy through two pathways: the thermal heat is utilized to drive a steam engine, while the produced hydrogen can be combusted in a separate chamber to generate high-temperature steam for powering an additional steam engine.

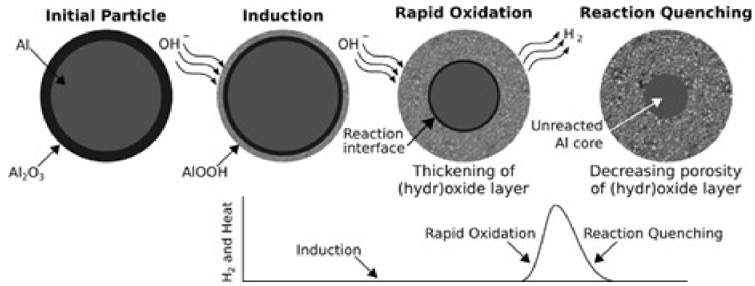


Figure 10: Schematic of the three reaction stages of aluminium with water at low temperatures, including a sketch of the initial dry particle for reference. Also sketched is the characteristic hydrogen-production curve showing the induction, rapid oxidation, and quenching stages [6]

Fig 10 illustrates the reaction pathway for the low temperature aluminium-water reaction, depicting an aluminium particle encased by a thin, inert aluminium oxide coating that must be overcome to facilitate metal-water oxidation. The metal-water reactor design requires careful optimization to achieve appropriate reaction rates for marine propulsion applications, while simultaneously maximizing hydrogen yield to enhance both the efficiency and mass energy density of the aluminium fuel.

The reaction rate and hydrogen yield can be enhanced through:

- Increased temperature and/or pressure conditions
- Reduced aluminium powder particle size
- Reduced aluminium powder to water ratio
- Implementation of various catalysts (detailed in following sub-section) [6, 4, 33]

1.3.1 Low Temperature Metal-Water Reactors

The aluminium-water reaction pathway for hydrogen generation has garnered significant international research attention, with most approaches focusing on nano-sized aluminium powder combined with catalysts at low temperatures. This approach involves reducing the aluminium powder to nano-sized particles and employing various catalytic methods such as alkaline solutions and ball milling the aluminium powder with other metals such as lithium and gallium, or salts such as NaCl. The primary aim of these methods is to increase the exposed surface area of the aluminium particles and prevent the protective layer of aluminium oxide from stopping or slowing down the reaction [34, 33]. This enables the metal-water reaction to take place below 120°C. Companies such as Found Energy are working to facilitate the metal-water reaction at room temperature and pressure [35].

This metal-water reactor design comes with both advantages and disadvantages. One advantage is that the reaction takes place quickly and enables the system to rapidly increase or decrease energy output to meet shifting load demands. However, this design also presents significant disadvantages:

- **High fuel production costs**: The production of nano-sized particles that also need to be ball milled with catalyst metals or salts requires an entire production chain and is likely to increase fuel costs substantially [36].

- **Elevated transport and handling costs**: The high reactivity of the nano-sized particles with water is likely to make transport and handling costs very high, as the metal reacts with water to produce hydrogen, which is a highly combustible gas and can independently contribute to climate change [37]. The higher production and handling costs for the fuel are likely to damage the economic case for this technology, since fuel costs constitute a large percentage of the operating costs for marine shipping, which is also a low-margin industry.

- **Low-grade heat output and reduced energy density**: As discussed in the previous sections, approximately half of the chemical potential energy of the aluminium metal is released as heat when reacted with water (as shown in equations 1 and 2). To maximise the round-trip efficiency of the metal-water reactor, this heat energy must be utilised for engine power. However, in these low-temperature reactors, the heat output is low-grade and unsuitable for use in traditional and high-efficiency steam engines [34]. The low-grade heat could be used by Stirling engines, but these engines have low power-to-weight ratios and have not been mass-produced so far like other heat engines such as diesel or gas turbines have [38]. The low-grade heat could also perhaps be upgraded using heat pumps and used in conjunction with Organic Rankine Cycle engines, but this increases the system complexity and the technology risks for the entire system [39]. The inability to utilise the low-grade heat also reduces the effective mass and volumetric energy density of the metal fuel by half.

Although the low-grade heat cannot be used for engine power, it can be used for other heating applications such as hot water production or indoor spas [40]. These applications are particularly useful in the yacht and cruise ship markets.

1.3.2 High Temperature Metal-Water Reactor

High-temperature metal-water reactors operate at 250–400°C and 15–25 MPa for optimal reaction rates [36]. These reactors can use micron-sized particles instead of nano-sized particles as in the case of low-temperature metal-water reactors. Although high-temperature metal-water reactors have garnered less research focus relative to low-temperature reactors, they come with some significant advantages, especially relating to marine shipping applications:

- **Lower fuel production costs**: Micron-sized particles are much cheaper and simpler to produce industrially compared to nano-sized particles [36, 4, 34].
- **Improved fuel handling and transport**: The micron-sized particles are also inert at room temperature and pressure, as opposed to nano-sized particles which can readily react with water at room temperature (depending on the catalytic process that is being used). Because of the fuel's inert nature, it makes them easier to transport and handle, which further improves the economic case for these reactors.

- **High-grade heat output and improved energy efficiency**: High-temperature reactors produce high-grade heat output in the form of a steam and hydrogen mixture. This high-grade heat can be used to power high-efficiency steam engines, which nearly doubles the round-trip energy efficiency of aluminium as a secondary energy carrier [34]. It also increases the mass and volumetric energy densities of the aluminium fuel as more useful energy can be created using the same kilogram of fuel.

Fuel Type	**Fuel Mass Density (kWh/kg)**	**Engine Efficiency**	**By-Product to Fuel Weight Ratio**	**Final Energy Density (kWh/kg)**
Al powder (ref.)	8.61	0.40 (ass.)	x	3.44
$Al(OH)_3$ by-product	8.61	0.40 (ass.)	2.89	1.19
AlOOH by-product	8.61	0.40 (ass.)	2.22	1.55
Al_2O_3 by-product	8.61	0.40 (ass.)	1.89	1.82

Table 1: Comparison of different $Al-H_2O$ by-product scenarios. Adapted from [2, 3, 4, 5, 6, 1]

Fuel Type	**Volume Density (kg/L)**	**Final Volumetric Energy Density (kWh/L)**
Al powder (ref.)	2.7	9.30
$Al(OH)_3$ by-product	2.42	2.88
AlOOH by-product	3.03	4.70
Al_2O_3 by-product	3.97	7.23

Table 2: Volumetric Energy Densities of different $Al-H_2O$ combustion scenarios. Adapted from [2, 3, 7, 8, 9, 10, 11]

- **Superior metal oxide by-product**: The aluminium-water reaction can produce three main by-products: $Al(OH)_3$, AlOOH, and Al_2O_3. The by-product produced can change the effective mass energy density of the fuel system. The lowest energy density happens with $Al(OH)_3$, while the highest is with Al_2O_3, as shown in Tables 1 and 2. In Fig 11 we can see that as temperature and pressure increase, the probability of producing Al_2O_3 instead of $Al(OH)_3$ increases, thus further increasing the effective mass and volumetric energy density of the aluminium fuel system. Al_2O_3 can also be used directly as the input in the Hall-Héroult process, as opposed to $Al(OH)_3$, which needs to be processed into Al_2O_3 first [40].

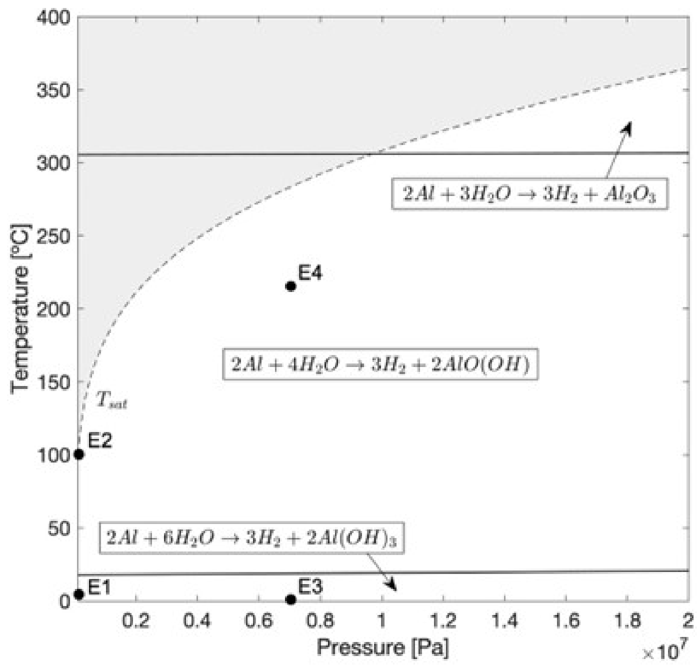


Figure 11: Al−H_2O reaction transition diagram extrapolated to 10 MPa. Labels E1-E4 show conditions used for experimental validation [18]

However, high-temperature reactors come with their own disadvantages:

- **Extended reactor start-up time**: It could take some time for the reactor to heat up, which can be up to 15–30 minutes depending on the size of the reactor. This problem can be mitigated by the use of hybrid electric propulsion using electrochemical batteries that work as a buffer storage system as the reactor heats up. Hybrid electric systems can mitigate the impact on performance and manoeuvrability caused by the longer start-up time. These hybrid electric systems have been widely used in the automotive sector for decades and have already been gaining traction in the marine shipping sector [41]. Hybrid electric designs were also used to manage the longer start-up time for aluminium powder powered submarine and UUV systems that were researched by the American military [32].

- **Higher reactor manufacturing costs**: The reactor for the high-temperature system is likely to be more expensive since it needs to handle the higher temperature and pressure.

1.3.3 Summary Tables

Parameter	LT-MWR	HT-MWR
Particle Size	Nano-sized particles	Micron-sized particles
Catalysts	Range of special alloys (Li, Ga, Hg, etc.)	No catalysts
Temperature and Pressure	Below 120°C (can operate at room temperature depending on catalyst)	250–400°C and 15–25 MPa
Metal By-product	$Al(OH)_3$ or AlOOH	AlOOH or Al_2O_3
Worst Case Mass Energy Density	0.60 or 0.78 kWh/kg	1.55 or 1.82 kWh/kg
Worst Case Volumetric Energy Density	1.44 or 2.35 kWh/L	4.70 or 7.23 kWh/L
Thermal Heat Utilization	Difficult due to low temperature	Possible with simple steam turbines
Fuel Production Costs	Higher (smaller particles and ball milling alloys)	Lower (larger particles and low catalysts)
Round-trip Energy Eff.	11–13% (if no heat is utilized)	23–25%
Fuel Transport Costs	Higher (very reactive at room temp.)	Lower (mostly inert at room temperature)
Fuel Handling Safety	Lower	Higher
Reactor Costs	Lower	Higher (to withstand high temperature & pressure)
Reactor Start-up Time	Lower (ability to load follow)	15–30 minutes (better for baseloads)

Table 3: Comparison of Low Temperature Metal-Water Reactors (LT-MWR) and High Temperature Metal-Water Reactors (HT-MWR)

1.4 Reactant Selection: Seawater versus Distilled Water

A key design decision is whether the reactant water comes from the ocean or is supplied as distilled water. While seawater appears attractive, historical precedent and certification pathways favour distilled water for a Generation-1 prototype.

Historical Precedent

Early steamships used seawater directly, but salt scaling and corrosion caused frequent failures as pressures rose. The surface condenser solved this: exhaust steam condensed back to distilled water for reuse. By the 1860s, designs like Alexander Chaplin & Co.'s were used on major lines. The British Admiralty formalised distilled feedwater in the 1880s, and G. & J. Weir built the first marine evaporator in 1884. By 1900, distilled water was standard – driven by reliability, not regulation.

Reactor Implications

Seawater introduces chlorides, sulphates, and biological matter that affect passivation, by-product composition, and fouling. For a Generation-1 system, distilled water offers three advantages:

1. **Simpler design** – predictable reactant enables closed-loop recycling (condensing H_2 exhaust), avoiding variable water quality.
2. **Lower technology risk** – isolates reactor from seawater chemistry, producing clean data comparable to prior McGill and Russian work.
3. **Easier certification** – classification societies prefer well-defined inputs for one-off builds (Section 1.6.3).

Research Prioritisation

The experimental programme (Section 3.2) prioritises distilled water as Tier 1 – the core needed for optimisation, by-product characterisation, and transient control. Seawater experiments are Tier 2 (synthetic) and Tier 3 (natural) – valuable but not required. Just as steamships shifted to distilled water for reliability, the first-generation reactor should follow the same path.

1.5 Economics of Aluminium Fuel

1.5.1 The Long Term Trends

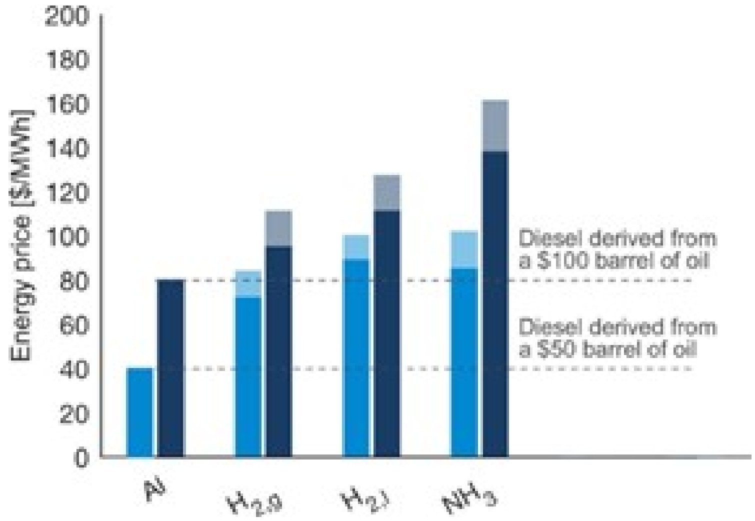


Figure 12: Aluminium fuel vs other electrofuels and diesel (The bars show the price of energy derived from low-carbon energy carriers, if the input electricity price were set to \$36/MWh (dark blue) or \$26/MWh (light blue). The pale region at the top of each bar represents the uncertainty related to the required inputs.) [1]

According to researchers at McGill University, Canada, if both the heat and hydrogen from the aluminium-water reaction are utilized for power generation at 40% efficiency, aluminium's round-trip efficiency as an electro-fuel reaches approximately 23-25%, making it comparable to hydrogen [42] and superior to ammonia [43].

The cost of electro-fuels like aluminium can be estimated based on electricity prices, while diesel is tied to crude oil prices. As shown in Figure 12, aluminium becomes cost-competitive with diesel when electricity prices fall between \$26-36/MWh and oil prices range from \$50-100 per barrel.

Aluminium also holds a cost advantage over rival electro-fuels such as hydrogen (both gas and liquefied forms) and ammonia. In OECD countries, industrial electricity rates range from \$45-143/MWh, with an average of \$100/MWh [44]. In regions with abundant renewable energy, such as Norway and Quebec, where rates are as low as \$45/MWh and \$27/MWh, aluminium becomes even more competitive in these markets [45].

1.5.2 The Emerging Green Aluminium Supply Chain

The economic case for aluminium as a recyclable energy carrier depends not only on electricity prices and round-trip efficiency, but also on the carbon intensity of aluminium production itself. For decades, the conventional Hall-Héroult process has been highly carbon-intensive for two fundamental reasons:

- **Very high electricity consumption** – Primary aluminium smelting typically requires 14–16 MWh of electricity per tonne of metal produced. In coal-dependent grids, this translates into a large carbon footprint.
- **Consumable carbon anodes** – The anodes used in conventional cells are made of baked carbon. During normal operation, these anodes oxidise to release carbon dioxide directly from the process, not just from the electricity source.

Together, these factors have made virgin aluminium one of the most emission-heavy industrial metals. Furthermore, the Hall-Héroult process was traditionally thought to require stable, inflexible baseload electricity. Aluminium smelters were designed to run continuously at a fixed power draw, making them a poor match for variable renewable electricity from solar and wind power.

However, several recent projects are challenging these long-held assumptions, pointing toward a future green aluminium supply chain that is both low-carbon and grid-interactive.

Photovoltaic direct-current (DC) aluminium smelting in Yunnan, China A novel DC-powered smelting system has been demonstrated that directly integrates photovoltaic generation with aluminium electrolysis. Key achievements include:

- DC power supply efficiency of at least **98%**, a significant improvement over the $\sim 80\%$ efficiency of traditional AC-coupled photovoltaic systems [46].
- Stabilisation of the electrolytic process by controlling current fluctuations on the DC bus to within **±1.5%**, overcoming a major technical barrier to direct solar-powered smelting [46].

This technology has already been deployed at **eight plant sites**, with a total installed capacity of **252 MW**, demonstrating that direct solar-powered aluminium production is scaling beyond the laboratory [47].

The EnPot process from New Zealand

The EnPot system provides, for the first time, dynamic control of the heat balance of aluminium smelting pots across an entire potline. Key capabilities include:

- Energy consumption and aluminium production can be increased or decreased by as much as **±30% almost instantaneously** [48].
- Smelters can reduce power draw during periods of grid stress or low renewable output, and increase draw when surplus renewable electricity is available.
- This transforms the smelter from a passive end-user of power into a **"virtual battery"** for the electricity network, enabling deeper penetration of wind and solar generation [48].

Crucially, the EnPot process is no longer a theoretical concept. It is being deployed in **real-world projects in Germany and China**, proving that flexible aluminium smelting can operate alongside heavy industrial demand and variable renewable grids [49].

Ceramic-based inert anodes

Perhaps the most transformative development is the ongoing commercialisation of inert anodes (ceramic or metal-based) to replace the century-old consumable carbon anodes. Inert anodes do not oxidise during electrolysis, eliminating the direct CO_2 emissions from the anode reaction. When paired with renewable electricity, the entire Hall-Héroult process becomes **completely zero-carbon** [30]. Several consortia (including Rusal, Alcoa, and Rio Tinto's ELYSIS partnership) are now scaling this technology [50].

Taken together, these three innovations – DC photovoltaic smelting, EnPot's load-flexible potlines, and inert anodes – are rapidly removing the historical carbon penalty from aluminium production. For the aluminium-water propulsion system proposed in this work, an emerging green aluminium supply chain means that the fuel can be truly low-carbon from well-to-wake, not only during onboard combustion. This strengthens both the environmental and economic case for aluminium as a marine fuel, particularly as renewable electricity becomes cheaper and more widespread.

1.6 Technology Development Pathway

1.6.1 Barriers to Entry in Container Shipping

The maritime shipping industry has historically been conservative in its adoption of new propulsion technologies. This reluctance stems from the sector's high capital costs and persistently low operating margins, which leave little room for experimental or unproven systems. Consequently, any novel engine or fuel system must first obtain certification from the International Maritime Organization (IMO), a process that typically requires extensive operational data and can take up to a decade. This lengthy and expensive approval pathway presents a formidable barrier to the introduction of a completely new fuel–engine combination, even in scenarios where high carbon taxes or stringent emissions regulations might otherwise favour low-carbon alternatives.

Given these challenges, a direct assault on the container shipping market with aluminium-water propulsion is unlikely to succeed in the near term. Moreover, while solar and battery costs are falling sharply, a robust green aluminium production industry has yet to emerge. Realistic projections suggest that at least 10–15 years will be required for this supply chain to mature and for aluminium fuel to reach cost parity with heavy fuel oil (HFO) and other hydrocarbons. Therefore, a beachhead market must be identified—one in which the technology can be developed, demonstrated, and refined through a "learning by doing" process, while generating the operational track record needed for eventual IMO certification. [51]

1.6.2 Luxury Yachts and Superyachts as an Ideal Beachhead

The most promising initial market for aluminium-water propulsion systems is the luxury yacht and superyacht segment. These vessels are typically purchased by Ultra-High-Net-Worth Individuals (UHNWIs), defined as individuals with a net worth of US$30 million or more. The global UHNWI population is expected to grow by approximately 33% between 2025 and 2030 [52], driving strong demand for bespoke, high-end vessels.

Unlike container shipping, which is a commoditised market driven almost exclusively by cost per tonne-mile, the superyacht market places a premium on product differentiation, innovation, and environmental stewardship. UHNWIs are also more likely to tolerate a significant green premium during the early years of technology development, while the aluminium-water propulsion system is being refined and while green aluminium fuel has yet to reach cost parity with conventional fuels. This tolerance for higher upfront costs provides a critical financial bridge for the technology.

Figure 13: Modern Electric Yachts with Integrated Solar Panels [19]

Figure 14: The Gotland Horizon X By Austal Panels [20]

Indeed, several green yachting options already exist, including fully battery-electric yachts powered by integrated solar panels (as shown in Fig 13). Furthermore, Austal—Australia's largest shipbuilder—is developing a catamaran ferry called the Gotland Horizon X (as shown in Fig 14), which features a lightweight "green aluminium" hull and can be powered entirely by green hydrogen [20]. These developments indicate a strong and growing appetite for low-carbon, novel vessel designs within the premium marine segment.

1.6.3 Regulatory Pathway and Certification Advantages

Equally important are the regulatory differences between commercial shipping and the superyacht sector. IMO certification for container shipping follows a rigid, top-down process that demands extensive pre-approved data. In contrast, superyacht certification is handled primarily by private classification societies such as Registro Italiano Navale (RINA), Lloyd's Register (LR), and DNV. These societies employ a flexible, bottom-up, risk-based approach, often engaging in bespoke negotiations with builders and owners. They are specifically equipped to approve novel systems and one-off builds, making them far more accommodating to innovative propulsion technologies [53, 54].

1.6.4 Pathway to Broader Adoption

If aluminium-water combustion systems can be successfully commercialised in the luxury yacht market, the operational data generated will greatly simplify the subsequent IMO certification process for container shipping applications. This strategy mirrors the path taken by Tesla in the automotive industry: the original Tesla Roadster—a high-performance, low-volume sports car—served as a beachhead to develop battery and electric drivetrain technology, generate real-world data, and build brand credibility before moving into mass-market vehicles [55, 51]. By targeting yachts and superyachts as the initial product development segment, the aluminium fuel system can follow an analogous learning-by-doing trajectory, ultimately paving the way for large-scale deployment in deep-sea shipping.

2 Research Questions

2.1 System Requirements for Yacht Applications

While deep-sea shipping remains the ultimate decarbonisation target, accounting for approximately 70% of the industry's fuel consumption and carbon emissions [27], the luxury yacht and superyacht segment offers an ideal beachhead for the development of aluminium-water propulsion systems. This market allows for iterative technology refinement, real-world operational data collection, and the establishment of flexible regulatory pathways.

However, the performance requirements for an aluminium-water reactor vary significantly across the different classes of superyachts that are popular in the market today. These include motor yachts, sailing superyachts, explorer yachts, and pocket superyachts (24–40 metres). To guide engineering design effectively, the following technical parameters must be systematically determined for each class:

- Minimum mass energy density
- Minimum volumetric energy density
- Power demand profile (minimum, maximum, and average power)
- Minimum range
- Total energy demand between refuelling stops
- Detailed energy demand profile for marine engine operations

2.2 Core Reactor Characterisation

Based on the analysis presented in Section 1.3, this research explicitly adopts the high-temperature metal-water reactor (HT-MWR) as the design architecture under investigation. The case for this choice over low-temperature alternatives rests on four compounding advantages:

1. **Lower fuel production costs.** The HT-MWR operates with micron-sized aluminium particles, which are substantially cheaper and simpler to produce industrially than the nano-sized particles required by low-temperature designs.

2. **Safer and cheaper fuel transport and handling.** Micron-sized particles are inert at room temperature and pressure, unlike nano-sized particles, which can react readily with ambient moisture. This inert character significantly reduces transport and handling costs and improves the overall economic case for aluminium as a marine fuel.

3. **Higher round-trip energy efficiency.** The HT-MWR produces high-grade thermal output—heat at temperatures suitable for use in steam engines and other external heat engines. Harnessing this thermal output alongside the hydrogen combustion effectively doubles the round-trip energy efficiency of the aluminium fuel cycle, and correspondingly increases its effective mass and volumetric energy density, as demonstrated in Section 1.3.

4. **Superior by-product for recycling.** At the elevated temperatures and pressures of the HT-MWR, the aluminium-water reaction preferentially produces Al_2O_3 rather than $Al(OH)_3$ or AlOOH. As shown in Tables 1 and 2, Al_2O_3 gives the most favourable effective energy density of any by-product scenario. Crucially, it can also be fed directly into the Hall–Héroult recycling process with minimal pre-processing, which reduces costs and complexity across the entire aluminium fuel supply chain.

Reactor Research Deliverables

The experimental programme is structured as a tiered investigation. The core distilled water experiments constitute Tier 1 and are essential for the project's success. These provide all data required for control system development and battery sizing. Synthetic and natural seawater experiments are designated as Tier 2 and Tier 3 respectively—they are valuable for understanding seawater's effects but are not required for the primary research objectives. This tiered structure ensures that the thesis remains defensible on distilled water experiments alone, while providing flexibility to extend the work if time and resources permit.

With this scope established, the research decomposes into the following primary questions:

1. Reaction parameter optimisation How do temperature (250–400°C), pressure (up to 25 MPa), aluminium particle size, and fuel-to-water ratio individually and jointly influence hydrogen yield and thermal heat output? The target thermal output is a steam-hydrogen mixture at 400–500°C—the temperature range required to drive external heat engines such as steam turbines. The goal of this investigation is to identify the combination of parameters that delivers stable, controllable output at this temperature while maximising hydrogen yield. As a secondary metric, the cold-start warm-up time from ambient to operating temperature will also be recorded for each optimal setpoint, to inform later battery sizing.

2. Aluminium particle size characterisation The reaction rate of the aluminium-water system increases as particle size decreases, but smaller particles are more expensive to produce and more difficult to handle safely. This investigation will therefore identify the coarsest particle size that still achieves the reaction rates and thermal output required to meet the power demand profile of the target superyacht class identified in Section 2.1, since this coarsest viable size defines the most commercially attractive operating point.

3. By-product composition analysis The by-product produced by the aluminium-water reaction—whether $Al(OH)_3$, AlOOH, or Al_2O_3—has a major impact on the practical performance of the fuel system, for two reasons. First, the yacht must carry the by-product in a dedicated storage tank throughout the voyage, so the mass and volume of the by-product directly affects the effective energy density available for propulsion, as shown in Tables 1 and 2. Second, delivering Al_2O_3 rather than $Al(OH)_3$ to recycling facilities eliminates an intermediate processing step before the Hall–Héroult process, reducing cost and complexity across the supply chain.

This investigation will therefore identify the specific temperature and pressure conditions under which the reaction shifts from producing $Al(OH)_3$ or AlOOH toward Al_2O_3 as the dominant by-product, and will characterise how fuel-to-water ratio and particle size interact with this shift.

4. Seawater reactant characterisation For a vessel operating at sea, seawater is the naturally available reactant. The ability to use seawater directly as the oxidiser would deliver significant operational and maintenance benefits for aluminium-powered marine vessels. However, seawater introduces chemical variables that distilled water experiments cannot capture:

- Dissolved chloride ions and sulphates, which may modify the passivation behaviour of the aluminium oxide layer and alter the induction, rapid oxidation, and quenching stages of the reaction
- Magnesium and sodium salts, which could affect by-product composition
- Biological matter and suspended particulates, which may affect reactor internals and feed systems over time

This investigation will compare the hydrogen yield and thermal output of the aluminium-water reaction using distilled water versus seawater. Any filtration or pre-treatment required for the seawater intake will be identified and documented, since this information is essential for a proper cost-benefit analysis of seawater as the operational oxidiser. Critically, this investigation is designated as Tier 2 —it will be pursued only if the core distilled water experiments are completed ahead of schedule. The thesis remains defensible on distilled water experiments alone.

5. Marine safety systems Aluminium powder introduces a set of hazards that are not commonly encountered in conventional marine fuel systems and are not fully addressed by existing maritime fuel codes:

- **Fire and explosion risk** from highly combustible aluminium dust clouds
- **Hydrogen liberation** from the exothermic reaction of aluminium powder with moisture, presenting both a fire risk and a potential contribution to climate change if vented
- **Chronic inhalation risk**—prolonged exposure to aluminium dust can cause pulmonary fibrosis (aluminosis)
- **Special extinguishing requirements**—water-based extinguishing agents are prohibited in areas where aluminium powder is present; Class D agents and inert atmospheres are required

The development and validation of safety protocols covering bunkering procedures, leak detection, dust control, and emergency response constitutes a distinct research deliverable of this study. These protocols will be developed in consultation with Registro Italiano Navale (RINA), Lloyd's Register (LR), and DNV, to ensure that the procedures adopted are consistent with the risk-based certification frameworks that will govern eventual onboard deployment, and that the documentation generated is structured to support future certification submissions.

Trade-offs and Design Decisions

As with any engineering system, the experimental results from this research will surface trade-offs that cannot all be optimised simultaneously. All such trade-offs will be transparently documented and justified as part of this study, providing future engineers with the information they need to make informed design decisions as the metal-water propulsion system is developed further toward commercial deployment.

2.3 Control Systems Development

While the experimental characterisation described in Section 2.2 will establish the steady-state and transient behaviour of the HT-MWR under open-loop conditions, a marine propulsion system requires closed-loop regulation. A superyacht's power demand is not constant; it changes continuously during harbour manoeuvring, coastal transits, and open-sea cruising. To be practically useful, the reactor must be capable of receiving a power setpoint and adjusting its output accordingly, while maintaining safe and stable operation.

However, a full continuous-flow reactor—with real-time fuel injection, dynamic slurry pumping, and continuous by-product extraction—introduces substantial mechanical complexity that is independent of the core thermal and kinetic control problem. Before attempting to regulate a continuous system, it is necessary to demonstrate that the reactor itself can be reliably stabilised at multiple power levels and transition between them with predictable dynamics. This research therefore scopes the control investigation to a *fixed-batch* configuration, in which a known mass of aluminium fuel is loaded at the start of a run and the reaction proceeds until depletion. This simplification isolates the control problem to its essential core—modulating the reaction rate via accessible process variables—while deferring the mechanical engineering of continuous fuel and solids handling to future work.

The control investigation addresses the following primary research questions:

1. **Transient response characteristics.** When a step change in power output is demanded between two operating levels, what are the measured response lag, rise time, settling time, and overshoot or undershoot in hydrogen production and thermal output? How do these dynamic metrics vary depending on the direction of the transition (e.g., increasing power versus decreasing power)?

2. **Stability and controllability limits.** Across the range of output levels required for marine propulsion, does the reactor exhibit stable behaviour, or do certain operating regimes induce oscillatory, sluggish, or runaway tendencies? What is the maximum rate of change of power demand that the reactor can track without violating safe operating limits (e.g., pressure or temperature excursions)?

3. **Comparison against operational requirements.** How do the measured transient response times and stability limits compare against the power demand profiles and manoeuvring requirements established for the target superyacht class in Section 3.1? This comparison determines whether the reactor, in its current configuration, can meet the dynamic performance expected of a marine propulsion system, or whether external buffering (such as battery storage) is essential to bridge the gap.

4. **Mode granularity optimisation.** The number and spacing of discrete power modes is a design choice, not a fixed constraint. What is the optimal mode granularity—for instance, three, five, or a continuously variable strategy—that balances controllability, manoeuvrability, and fuel utilisation? This investigation will assess whether a coarse discretisation is sufficient for smooth propulsion, or whether finer resolution is required to avoid unacceptable lags or instability.

Critically, this research does *not* design the battery buffer system that would compensate for reactor start-up and load-following limitations—that task is identified as future work in Table 4. However, the dynamic characterisation produced here—the measured lag times, ramp rates, and stability boundaries—constitutes the primary quantitative input that future engineers will require to correctly size that battery buffer, select its chemistry, and specify its power electronics. By delivering a validated closed-loop characterisation of the HT-MWR, this research transforms the reactor from a laboratory-scale demonstrator into an engineering subsystem with known, documented performance boundaries.

2.4 Engine and Battery System Analysis

The HT-MWR produces a high-temperature, high-pressure output stream composed of steam and hydrogen gas. This output must ultimately be converted into mechanical shaft power to drive a propeller. The reactor research in Sections 2.2 and 2.3 establishes the thermodynamic state of that output—its temperature, pressure, mass flow rate, and hydrogen fraction—but it does not prescribe how that output should be transformed into propulsion. The choice of power conversion architecture is therefore a critical design decision that affects efficiency, weight, footprint, capital cost, and—most importantly for near-term deployment—regulatory certification risk.

The analysis in this section is explicitly scoped as a *requirements-definition and trade-off study*, not an engineering design. The goal is not to design a specific engine or battery system, but rather to survey the available options, apply a consistent set of selection criteria, and produce a clear, defensible recommendation for the most viable Generation-1 propulsion architecture. Detailed simulation, integration, and prototype testing of the engine and battery systems are explicitly out of scope and are identified as future work in Table 4.

The analysis is guided by a single overarching principle: **minimise technical and certification risk for the metal-water reactor and aluminium fuel system**. The reactor and its fuel cycle are the novel, unproven elements of the propulsion chain. The engine and battery, by contrast, should be as mature, standardised, and readily certifiable as possible. Any complexity or novelty allocated to the prime mover or energy storage subsystem increases the overall system risk and lengthens the pathway to regulatory approval. The analysis therefore deliberately favours components with existing marine service records, established classification society rules, and straightforward integration pathways, even if this means accepting lower thermal efficiency.

The engine and battery analysis addresses the following primary research questions:

1. **Optimal thermal integration pathway.** The steam-hydrogen mixture can be exploited either by direct thermal combustion (burning the hydrogen to superheat the steam for turbine expansion) or by condensing the steam and separating the hydrogen for use in a separate prime mover. Which pathway—direct thermal integration or hydrogen separation—minimises overall system complexity, weight, and certification risk while still meeting the performance requirements of the target superyacht class?

2. **Prime mover selection and interface specification.** Among commercially available, mature marine prime movers suitable for the selected thermal integration pathway, which offers the best combination of technical maturity, certification feasibility, and efficiency for a Generation-1 prototype? What are the critical interface requirements—steam temperature, pressure, flow rate, hydrogen tolerance, and transient ramp-rate limits—that the reactor must deliver to interface with that prime mover? (Speculative or developmental prime movers with limited marine certification will be explicitly deprioritised.)

3. **Battery buffer specification.** Given the quantified reactor start-up and transient lag times from Section 2.3, what are the minimum energy capacity (kWh) and power rating (kW) that a buffer battery must provide to ensure uninterrupted propulsion during mode transitions? Which commercially available marine battery chemistries and form factors are best suited to this duty cycle, and what is the simplest hybrid architecture (e.g., series, parallel, or combined) that meets the buffering requirement with the lowest system risk?

4. **Efficiency target relaxation.** The preliminary analysis in Section 1.1 assumed a 40% engine efficiency for metal fuels, a value derived from theoretical projections rather than real-world demonstration. For a Generation-1 prototype, what is a realistic, defensible efficiency target that balances technical feasibility against commercial viability? More specifically, can the system achieve acceptable performance and economic competitiveness if the prime mover efficiency is relaxed to the 20–30% range, which is more readily attainable with mature, off-the-shelf components?

The outputs of this analysis will be a set of **engineering specifications**—precise interface requirements between the reactor, the prime mover, and the battery buffer—and a **clear architectural recommendation** for the minimum viable Generation-1 propulsion system. By defining these interfaces in terms of commercially available, certifiable components, the analysis reduces the overall technology risk and provides a concrete, actionable roadmap for the next phase of prototype development.

2.5 Scope of This Research

Research Activity	In Scope	Out Scope	How This Study Enables Future Work
Core Reactor Characterisation			
Industry requirements matrix	Yes	—	Defines power, range, and energy density targets that bound all downstream reactor design
Reaction parameter characterisation (T, P, particle size, fuel-to-water ratio)	Yes (Tier 1)	—	Produces the experimental dataset for reactor geometry and operating point selection
By-product composition analysis ($Al(OH)_3$, AlOOH, Al_2O_3)	Yes (Tier 1)	—	Determines achievable effective energy density and recycling compatibility under real operating conditions
Seawater vs. distilled water reactant comparison	Yes (Tier 2)	—	Establishes whether onboard seawater can be used directly, and specifies any pre-treatment or filtration requirements
Reactor start-up time and transient load characterisation	Yes (Tier 1)	—	Provides the open-loop dynamic data required for control system design and battery buffer specification
Marine safety protocols (bunkering, dust control, emergency response)	Yes	—	Provides the safety evidential basis for future certification submissions to RINA, LR, and DNV
Reactor geometry and material selection	Yes (lab-scale, indicative)	Full-scale design	Lab-scale findings provide the basis for scaled engineering design in future work
Preliminary cost estimation (reactor and aluminium fuel)	Yes (indicative, order-of-magnitude)	Final production costs	High-level estimates inform the economic feasibility case and identify where further cost refinement is needed
Control Systems			

Continued on next page...

Research Activity	In Scope	Out Scope	How This Study Enables Future Work
Closed-loop control system development	Yes (fixed-batch demonstrator)	Production-ready control hardware	Demonstrates thermal and kinetic controllability; quantifies response lags, rise times, and stability boundaries
Microcontroller hardware selection and implementation	Yes	Maritime ruggedisation	Provides a validated testbed for control algorithm development; hardware choices inform future procurement specifications
Power mode transition characterisation	Yes	Optimal controller design	Quantifies the reactor's dynamic response to step changes, enabling future engineers to size battery buffers and design control systems
Mode granularity optimisation	Yes	—	Determines the optimal number and spacing of discrete power modes for marine propulsion, balancing controllability and manoeuvrability
Engine and Battery Analysis			
Thermal integration pathway trade-off (direct vs. separation)	Yes	Detailed separation system design	Identifies the lowest-risk pathway for converting the steam-H_2 mixture into shaft power
Prime mover survey and selection	Yes (requirements definition)	Engine design or integration	Produces a decision matrix comparing mature, certifiable prime movers (steam turbine, ORC, diesel)
Battery buffer specification	Yes (requirements definition)	Battery system design	Defines minimum energy (kWh) and power (kW) requirements for a marine-certified battery to bridge reactor transients
Hybrid architecture recommendation	Yes (requirements definition)	Hybrid system integration	Recommends the simplest hybrid topology (series, parallel, or combined) that meets buffering requirements with lowest system risk
Efficiency target relaxation analysis	Yes	Efficiency optimisation	Establishes realistic, defensible efficiency targets (20–30%) for a Generation-1 prototype
Out of Scope (Future Work)			

Continued on next page...

Research Activity	In Scope	Out Scope	How This Study Enables Future Work
Full-scale reactor design and stress testing	—	Yes	Lab-scale findings from this study provide the basis for scaling engineering design in future work
Continuous-flow reactor (real-time fuel injection & solids handling)	—	Yes	Fixed-batch control data demonstrates fundamental controllability; continuous-flow engineering is a separate mechanical design challenge
Engine and turbine detailed design and integration	—	Yes	Interface specifications (steam temperature, pressure, flow rate) defined here enable future procurement and integration
Battery system detailed design and BMS development	—	Yes	Capacity and power requirements defined here enable future procurement and integration
Classification society approval (full-scale)	—	Yes	Safety protocols and experimental datasets developed here form the evidential basis for future certification submissions

3 Methodology

3.1 Industry Research

Research Design

The industry research component of this study seeks to establish the quantitative system requirements and qualitative market context that will bound the subsequent reactor design work. The information required spans two distinct domains: technical parameters such as power demand profiles, energy density targets, range requirements, and refuelling constraints; and contextual parameters such as risk tolerance, segment-specific adoption barriers, and willingness to pay a green premium. No validated dataset exists for either domain in the context of aluminium–water propulsion for superyachts, since the technology itself is novel. A fixed-format survey instrument is therefore inappropriate at this stage — the questions themselves require contextual interpretation and the ability to pursue unexpectedly informative responses in depth, neither of which a questionnaire can accommodate.

Semi-structured interviews are the appropriate method for this type of exploratory technical and commercial research. They ensure that a consistent set of core topics is covered across all participants — enabling structured comparison and synthesis — while preserving the flexibility to probe unexpected responses and allow participants to introduce considerations the researcher had not anticipated [56]. This flexibility is especially important when interviewing across multiple stakeholder groups with different professional languages and priorities, as is the case here.

Stakeholder Groups and Sampling Strategy

Participants will be drawn from four groups, each contributing a different category of information:

1. **Superyacht builders and shipyards** will provide technical data on vessel power systems, structural constraints, and storage configurations across the target vessel classes.

2. **Classification societies** — specifically RINA, Lloyd's Register, and DNV — will clarify the regulatory pathway for novel propulsion systems, documentation requirements for prototype approval, and specific concerns regarding aluminium powder handling and onboard hydrogen production.

3. **Naval architects** will supply quantitative power demand profiles, range calculations, and energy density constraints for specific vessel classes with a precision that neither builders nor owners can typically provide.

4. **Vessel owners and superyacht management companies** will contribute priorities around operational flexibility, risk tolerance, and the green premium threshold above which an aluminium–water system becomes commercially unacceptable.

Participants within each group will be identified through purposive sampling — a deliberate, non-random selection strategy that targets individuals with specific expertise, roles, or institutional knowledge relevant to the research questions [57]. This is the appropriate sampling strategy when the goal is depth of insight rather than statistical representativeness. Initial contacts will be identified through industry directories, professional networks, and direct outreach to relevant organisations. A snowball sampling component will supplement this within each group: each participant will be asked to suggest other relevant contacts, providing access to practitioners who may not be publicly visible through standard directories. Key industry events — including the Monaco Yacht Show, the Superyacht Forum, and Europort — will be attended where feasible to facilitate direct introductions to potential participants.

The target sample size is four to six participants per stakeholder group, giving a total of 16–24 interviews across the study. Research on thematic saturation in expert interview studies indicates that the majority of core themes are typically identifiable within six interviews in a relatively homogeneous professional population, and that saturation is rarely reached beyond twelve [58]. Since each of the four groups is well-defined in terms of knowledge domain and professional role, a target of four to six per group provides a reasonable expectation of saturation within the first research quarter.

Interview Guide Development

A separate semi-structured interview guide will be developed for each of the four stakeholder groups. Each guide will share a common core covering the topics essential to the requirements matrix but will be adapted in language, technical emphasis, and sequencing to suit each group. The core topics across all guides are:

- Typical energy demand profiles and power output requirements for the target vessel classes, across distinct operational phases including docking, harbour manoeuvring, coastal transit, and offshore cruising
- Total energy demand between planned refuelling stops, and the maximum acceptable refuelling time and downtime per stop
- Onboard space and weight constraints affecting fuel storage, by-product storage, and propulsion system footprint
- Tolerance for novel technology risk and willingness to pay a green premium during the early commercialisation phase of the technology

- Views on the regulatory and certification pathway for a novel propulsion fuel system

The classification society guide will additionally address the documentation required for prototype approval on a one-off build, the operational data needed to support progression toward type approval for series production, and specific regulatory concerns related to aluminium powder bunkering and onboard hydrogen management.

Before deployment with research participants, the guide will be piloted with two individuals who will not be included in the final analytical sample — ideally academics or industry consultants with superyacht or maritime propulsion experience. Piloting serves to identify ambiguous or leading questions, test the logical flow of the session, and calibrate the time required. The guide will be revised based on pilot feedback before the main data collection begins.

Data Collection

All interviews will be conducted by the principal researcher to ensure consistency of facilitation. Where possible, interviews will be held in person at the participant's place of work, as the professional environment can itself provide contextual information and facilitates rapport. Where in-person meetings are impractical due to geography or scheduling, video-conferencing will be used. Each session will target 45–60 minutes.

Before each interview, participants will receive a written information sheet describing the research purpose, the voluntary nature of participation, the intended use of the data, and their right to withdraw at any time without consequence. Written informed consent will be obtained before the session begins. With consent, all interviews will be audio-recorded and subsequently transcribed verbatim. Where recording is declined — which may occur with some classification society representatives given institutional confidentiality obligations — detailed contemporaneous notes will be taken during the session and written up in full within 24 hours of the interview. All data will be stored in accordance with the university's research data management policy, with participant identities anonymised in all transcripts and analytical outputs. Organisations may be identified by type (e.g. "a major European classification society") but not by name in any published output, unless explicit permission is granted.

Data Analysis

Transcripts will be analysed using reflexive thematic analysis, following the six-phase process described by Braun and Clarke [59]: familiarisation with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the final written analysis. This approach is appropriate because the research is fundamentally exploratory — the goal is to identify patterns and points of convergence or divergence across stakeholder perspectives, rather than to test a pre-specified hypothesis.

Two distinct types of output will be extracted from the analysis. First, quantitative parameters — power demand values, range requirements, refuelling time tolerances, storage volume constraints, and energy density targets — will be drawn directly from interview content, tabulated by vessel class, and where multiple participants provide estimates, cross-validated and reconciled to produce working values with documented uncertainty ranges. Second, qualitative themes — covering risk tolerance, green premium thresholds, segment-specific adoption barriers, and regulatory pathway insights — will be synthesised across participant groups, with representative quotations used to illustrate key findings.

To strengthen the credibility of the analysis, member checking will be conducted once the initial thematic analysis is complete [60]. A summary of the key findings and the draft requirements matrix will be shared with at least one participant per stakeholder group for comment and correction before the matrix is finalised. This practice reduces the risk of misinterpretation and ensures that the technical parameters derived from the interviews are recognised as accurate by the domain experts who provided them.

Synthesis into the Requirements Matrix

The outputs of the thematic analysis will be synthesised into a formal requirements matrix — the primary deliverable of the industry research phase. This matrix will document the minimum and maximum values for each technical parameter for each superyacht class investigated, alongside the qualitative constraints and stakeholder priorities identified across the four groups.

The requirements matrix serves two functions within the broader research project. First, it identifies which superyacht class or classes represent the most promising initial deployment target for the aluminium–water propulsion system, based on the combination of regulatory flexibility, risk tolerance, and alignment between operational requirements and the known performance envelope of the HT-MWR. Second, it directly bounds the experimental programme in Section 3.2: the power output targets for the transient load characterisation, the energy density thresholds for the parameter optimisation experiments, and the range requirements constraining by-product storage calculations will all be set by the values documented in this matrix.

3.2 Core Reactor Characterisation

This section describes the experimental programme for the high-temperature metal-water reactor (HT-MWR) component of this research. The programme is structured to address the steady-state research deliverables identified in Section 2.2, building on the system requirements established through the industry research in Section 3.1.

Reactor Vessel and Instrumentation

The central piece of experimental equipment is a laboratory-scale high-pressure reactor vessel designed to operate at temperatures of 250–400°C and pressures up to 25 MPa. The reactor body will be constructed from SS316L stainless steel, which provides the required tensile strength, corrosion resistance in high-temperature steam environments, and wide commercial availability of pressure-rated fittings. All pressurised components will be specified with pressure ratings exceeding 300 bar at 450°C. For seawater experiments, reactor sections in contact with the fluid will be constructed from or lined with Hastelloy C-276 or Inconel 625 to resist chloride-induced stress corrosion cracking. The reactor will be heated externally using a resistance heater, with thermal insulation applied along its full length to minimise heat loss.

The following instrumentation will be installed as standard across all experiments:

- **Temperature**: Four K-type thermocouples distributed axially along the reactor exterior, with the average reading used as the representative reactor temperature, consistent with the approach of Vlaskin et al. [36]

- **Pressure**: Two calibrated high-pressure transducers — one on the pump inlet and one measuring direct reactor pressure — with continuous high-frequency data logging

- **Real-time gas flow**: A thermal mass flow controller on the gas outlet line, providing continuous volumetric measurement of gas leaving the reactor

- **Hydrogen concentration**: An electrochemical hydrogen sensor positioned in the gas outlet stream and exhaust ventilation line, serving both as a safety monitor and a supplementary yield measurement

- **Slurry injection**: A variable-speed high-pressure plunger pump with remote speed control, enabling precise and adjustable mass flow rates of aluminium–water slurry into the reactor

All sensor outputs will be logged centrally from a remote data acquisition console outside the reactor enclosure.

Aluminium Powder Characterisation

Seven discrete particle size grades will be tested, spanning the 10–500 micrometre range, with target nominal sizes of approximately 10, 25, 50, 100, 200, 350, and 500 µm. These will be selected from commercially available aluminium powder products, covering the size range studied in prior high-temperature aluminium–water research in Russia [36] and Canada [61]. Before use in any reaction experiment, each powder batch will be characterised using:

- **Laser diffraction particle size analysis**: To verify actual size distributions (D10, D50, and D90 values) against nominal supplier specifications
- **Scanning electron microscopy (SEM)**: To inspect particle morphology, surface condition, and the native aluminium oxide passivation layer
- **Inductively coupled plasma optical emission spectroscopy (ICP-OES)**: To quantify elemental impurities, particularly silicon and iron content

Silicon content will be screened carefully, as Boudreau et al. [61] demonstrated that increasing silicon mass fraction from 0.06% to 0.54% reduced aluminium oxidation rate by an order of magnitude under supercritical water conditions. Only powders with silicon content below 0.1% by mass will be used in the primary parameter optimisation experiments.

Reaction Parameter Optimisation

Experiments will systematically vary temperature (250–400°C), pressure (saturated vapour pressure at each temperature, with supplementary runs up to 25 MPa), water-to-aluminium mass ratio (5, 7, 9, and 12), and all seven particle size grades. Each parameter combination will be repeated a minimum of three times to establish repeatability. The primary measured outputs are:

1. **Hydrogen yield**: Calculated from reactor pressure rise using the ideal gas law and cross-validated against the mass flow controller reading on the gas outlet
2. **Thermal heat output**: Derived from the reactor temperature–time curves recorded by the four thermocouples

The target output condition is a steam–hydrogen mixture at 400–500°C. The coarsest particle size that still achieves this target while meeting the power demand requirements of the superyacht class from Section 3.1 will be identified as the most commercially attractive operating point. The cold-start warm-up time from ambient to operating temperature will also be recorded for each optimal setpoint.

By-product Composition Analysis

Following each experiment, solid products will be extracted, cooled, and vacuum-dried before analysis using two complementary methods:

- **Powder X-ray diffraction (PXRD)**: To identify and quantify the crystalline phases present — specifically $Al(OH)_3$, AlOOH, and the Al_2O_3 polymorphs — with phase proportions quantified by Rietveld refinement of the diffraction patterns
- **Thermogravimetric analysis (TGA)**: As a complementary method to cross-validate phase composition based on characteristic mass loss temperatures

Residual unreacted aluminium will also be quantified by PXRD, providing a direct conversion degree measurement alongside the hydrogen yield calculations. By-product phase data will be mapped against the temperature and pressure conditions of each experiment to identify the threshold conditions required to shift the reaction toward Al_2O_3 as the dominant product.

Seawater Reactant Characterisation

Rather than a simple binary comparison between distilled water and seawater, this research defines four distinct water treatment categories to isolate which components of seawater affect the reaction:

- **Category A — Distilled water**: The baseline case
- **Category B — Synthetic seawater**: Distilled water with dissolved salts at open-ocean concentration (≈35 g/L, ASTM D1141), isolating ionic effects from biological and particulate contamination
- **Category C — Coarse-filtered natural seawater**: Natural seawater passed through a 50 µm filter, removing organisms and macro-particulates while retaining dissolved ions and fine particles
- **Category D — Fine-filtered natural seawater**: Natural seawater passed through a 50 µm pre-filter followed by a 1 µm membrane filter, removing virtually all suspended particulates while retaining dissolved ions

Natural seawater samples for Categories C and D will be collected at both open-ocean (≈35 g/L) and coastal (≈15 g/L) salinities. The hydrogen yield, thermal output, and by-product composition for each water type will be compared against the distilled water baseline. The filtration complexity and estimated cost of each category will be documented, enabling a cost-benefit analysis of onboard seawater pre-treatment for a deployed marine system.

Safety, Verification, and Approval

The experimental programme involves concurrent hazards from aluminium powder handling, continuous hydrogen gas production, and high-pressure and high-temperature reactor operation. A comprehensive safety protocol addressing each risk category, together with the plan for independent peer review and formal university approval of the experimental design, is provided in full in the Appendix. In summary, all experimental work will be conducted in a dedicated ventilated enclosure with Class D fire suppression, continuous hydrogen monitoring with automatic equipment interlocks, and a minimum of two researchers present at all times. A detailed laboratory experimental proposal incorporating the full safety protocol, risk register, and reactor specifications will be submitted for independent review by experienced academics and for formal approval by the university's laboratory safety committee. No high-pressure or high-temperature experimental work will commence before this approval is granted. This comprehensive proposal and its approval will be completed by the end of the first quarter of the research project.

3.3 Metal-Water Reactor Control Systems

Overview

The steady-state characterisation described in Section 3.2 identifies the optimal operating points — temperature, pressure, particle size, and fuel-to-water ratio — that maximise hydrogen yield and thermal output. A marine propulsion system, however, requires dynamic load-following: the reactor must accept a commanded power setpoint and adjust its output automatically while remaining stable and safe. This section describes the methodology for characterising the reactor's dynamic behaviour and developing a closed-loop control system for the fixed-batch reactor configuration introduced in Section 2.3.

Experimental Configuration: Fixed-Batch Reactor

For the control validation experiments, the reactor is operated in a *fixed-batch mode*. A known mass of aluminium powder (determined from the requirements matrix in Section 3.1) is loaded into the reactor at the start of each run. The reactor is then sealed, purged with nitrogen, and brought to thermal standby temperature. The reaction proceeds until the aluminium is fully consumed. This configuration deliberately excludes real-time fuel injection and continuous by-product extraction, which are deferred to future work. The simplification allows the control investigation to focus purely on thermal and kinetic regulation—the core intellectual challenge—without the confounding mechanical complexities of slurry pumping, nozzle clogging, and solids handling.

Power Mode Definition

The reactor's output will be regulated across **five discrete power modes**, defined as percentages of the rated output determined from the requirements matrix (Section 3.1):

- **0%** — Thermal standby. The reactor is maintained at a minimum temperature (e.g., 250°C) via electrical heater input, but the aluminium-water reaction is suppressed (water injection stopped). This preserves the ability to respond rapidly when higher power is demanded.
- **25%** — Corresponds to harbour manoeuvring, low-speed transit, or docking.
- **50%** — Corresponds to moderate cruising.
- **75%** — Corresponds to high-speed cruising.
- **100%** — Full rated output.

The absolute power targets (in kW thermal and kg H_2/hr) for each mode will be derived from the superyacht power demand profiles established in Section 3.1. The control system's task is to transition between these modes on command, maintaining each mode stably, and to characterise the dynamic behaviour of each transition.

Open-Loop Plant Characterisation (Prerequisite)

Before feedback control can be implemented, the reactor's inherent dynamic response must be quantified without any automated regulation. Using the reactor apparatus described in Section 3.2 — the same thermocouples, pressure transducers, mass flow controller, and hydrogen sensor — step-change tests are performed across the five power modes. Hydrogen production rate is calculated via the pressure-time derivative method (dP/dt) and cross-validated against the mass flow controller reading. For each transition (e.g., 25% to 75%), the following metrics are recorded:

- **Response lag** — Time from the step command to the first detectable change in output.
- **Rise time** — Time from first output change to reaching 90% of the final steady-state value.
- **Settling time** — Time to reach and remain within ±5% of the final steady-state value.
- **Overshoot / undershoot** — Maximum excursion above or below the target value, expressed as a percentage of the target.

- **Steady-state error** — Residual difference between the setpoint and the measured output after settling.

Each pairwise transition (0→25, 0→50, 0→75, 0→100, 25→50, 25→75, 25→100, 50→75, 50→100, 75→100) is repeated at least three times to assess repeatability. The direction of the transition (e.g., 25% to 75% vs. 75% to 25%) is treated separately, as the reactor may exhibit asymmetric behaviour.

This open-loop dataset establishes the raw plant dynamics — the gain, time constant, and dead-time that any future controller must handle. It also provides the baseline against which all closed-loop improvements will be measured.

Actuator Selection

The open-loop data informs the choice of primary control actuator. The reactor has two variables that can be modulated in real-time:

- **Water injection rate**: Controlled via the variable-speed high-pressure plunger pump. The open-loop step tests quantify the gain (change in output per unit change in pump speed), time constant, and linearity of this actuator.
- **Reactor temperature**: Modulated via the electrical resistance heater. Due to thermal inertia, this method has a slow response time — on the order of minutes rather than seconds.

The actuator with the faster, more linear response and shorter time constant will be selected as the primary means of load-following. The slower actuator (temperature) will be investigated as a secondary, coarse-trim actuator for steady-state fine-tuning or thermal standby management. If pump speed modulation alone proves insufficient to meet the transient response requirements from the requirements matrix (Section 3.1), a cascaded strategy may be used — fast adjustments via pump speed with slow temperature trim to compensate for drift as the aluminium is consumed.

Closed-Loop Controller Implementation

A closed-loop controller is implemented on a **microcontroller platform**—either a Raspberry Pi (for higher computational flexibility) or an Arduino-based system (for deterministic, low-latency real-time control). The final choice will be made during the design phase based on required sampling rate, control algorithm complexity, and I/O availability.

Hardware components:

- Microcontroller board (Raspberry Pi 4 or Arduino Due / Teensy 4.0)
- Analogue-to-digital converters (ADCs) for thermocouple and pressure transducer signals

- Digital-to-analogue converters (DACs) or PWM outputs for actuator control
- Signal conditioning boards (thermocouple amplifiers, pressure transducer excitations)
- Power relays / solid-state switches for heater control
- Motor driver / variable-speed drive for the plunger pump
- Isolated communication links (USB, Ethernet, or CAN) to the data acquisition console

The control algorithm itself is **not** a primary research deliverable. The objective is not to design an optimal controller, but to characterise the plant dynamics that any future controller would need to handle. Therefore, a simple **proportional-integral-derivative (PID)** controller will be implemented initially, with gains tuned using the gain and time constant extracted from the open-loop tests. If the plant exhibits significant non-linearity or time-varying behaviour (e.g., as the aluminium is consumed), a gain-scheduled or model-predictive controller may be explored. In all cases, the control scheme will be documented transparently, and the primary data product will be the measured closed-loop response.

Closed-Loop Step-Change Validation

The same step-change test matrix used in the open-loop characterisation is repeated under PID regulation. For each transition, the closed-loop response metrics — lag, rise time, settling time, overshoot, and steady-state error — are recorded and compared directly against the open-loop baselines to quantify the improvement.

Stability and Controllability Limits

To probe the stability envelope, the controller will be subjected to:

- **Large step changes** (e.g., 0% to 100% directly) to determine if the reactor can handle extreme transients without thermal runaway or pressure exceedance.
- **Ramp changes** (linear increase from 25% to 75% over 30, 60, and 120 seconds) to characterise the maximum tolerable slew rate.
- **Repeated cycling** between two modes (e.g., 25% to 75% repeated 5 times) to test for hysteresis or cumulative drift.

In the fixed-batch configuration, the progressive consumption of aluminium and accumulation of by-product introduce time-varying plant dynamics; the stability investigation explicitly characterises how controllability evolves over the course of a batch.

Mode Granularity Optimisation

The choice of five discrete modes is a design assumption, not a fixed constraint. The data from the open-loop and closed-loop transition characterisation will be used to evaluate whether five modes are appropriate. If the reactor can transition smoothly between widely spaced modes (e.g., 25% to 75%) with acceptable lags and overshoot, a **coarser discretisation** (e.g., 0%, 50%, 100%) may be sufficient, simplifying the control system. Conversely, if transitions between distant modes induce instability or unacceptable lag, a **finer discretisation** or a **continuously variable** strategy may be required. The recommendation will be grounded in quantitative data and will explicitly address the trade-off between control complexity and manoeuvrability performance.

Data Analysis and Comparison Against Requirements

The measured transition metrics — both open-loop and closed-loop — will be compared directly against the superyacht power demand profiles and manoeuvring requirements documented in the requirements matrix (Section 3.1). The key comparison is whether, for each power mode transition demanded by a typical sailing profile (e.g., harbour to cruising to docking), the reactor's measured response lag and rise time fall within the acceptable limits defined by the naval architects and owners.

If the reactor's response is faster than required, the control system can be simplified (e.g., slower, smoother ramps to reduce wear). If the response is slower than required, the data will quantify exactly how much battery buffering (energy and power) is needed to bridge the gap. Critically, the open-loop data quantifies the minimum required buffer, while the closed-loop data quantifies the achievable performance with control — together, they bound the battery sizing problem.

Limitations and Scope Boundary

This methodology is explicitly limited to **fixed-batch, laboratory-scale** validation. The control system is a **demonstrator**, not a production-ready marine system. The microcontroller hardware and software are chosen for ease of development and data logging, not for maritime ruggedisation or certification. The control algorithm is simplified (PID) and is not intended to be directly transferable to a full-scale vessel. However, the **dynamic response data** produced here — the transfer functions, time constants, and stability boundaries — is independent of the specific controller hardware. It constitutes the fundamental input that future engineers will use to design a ruggedised, certified control system for commercial deployment.

3.4 Engine and Battery System Analysis

Overview

This section describes the methodology for the systematic, comparative analysis of candidate engine and battery architectures for the Generation-1 aluminium-water propulsion system. Unlike the experimental components in Sections 3.2 and 3.3, this analysis is a **literature-based and manufacturer-data-based trade-off study**. It does not involve laboratory work, prototyping, or simulation. The objective is to produce a clear, well-justified architectural recommendation and a set of engineering interface specifications that will guide future prototype development. This is consistent with the scope boundary defined in Table 4, which explicitly identifies engine integration and battery design as future work.

The analysis is guided by the principle of **minimising technical and certification risk** for the aluminium-water reactor and fuel system. The engine and battery should be as mature, off-the-shelf, and readily certifiable as possible.

Step 1 — Definition of Evaluation Criteria

Before comparing any specific technologies, a set of quantitative and qualitative criteria will be defined. These criteria will be used to score each candidate architecture consistently. The criteria are:

1. **Technical maturity** — Has the technology been deployed in marine or adjacent industrial applications for more than 5 years? (Weighted high)
2. **Certification pathway** — Do classification societies (RINA, LR, DNV) have existing rules and guidance? (Weighted highest)
3. **Thermal efficiency** — What is the expected efficiency when operating on the reactor's steam-hydrogen mixture? (Gen-1 target: 20–30%) (Weighted medium)
4. **Power-to-weight ratio** — Is the engine suitable for the weight-constrained environment of a superyacht? (Weighted medium)
5. **Footprint and packaging** — Does the engine fit within the available machinery space? (Weighted low)
6. **Capital cost** — What is the approximate per-kW cost for a unit in the 100–500 kW range? (Weighted low)
7. **Maintenance complexity** — How frequent and costly are scheduled overhauls? (Weighted low)

8. **Sensitivity to hydrogen content** — Can the engine tolerate variations in the steam-H_2 mixture composition? (Weighted high)

9. **Integration effort** — How many additional subsystems are required to interface with the reactor? (Weighted high)

Each criterion will be scored on a 1–5 scale (5 being best), and a weighted sum will be calculated. The weighting factors will be justified explicitly in the thesis, with reference to the industry interview findings (Section 3.1).

Step 2 — Thermal Integration Pathway Trade-Off

The reactor output is a hot mixture of steam and hydrogen gas. Two fundamentally different architectures for converting this output into shaft power will be evaluated:

Option A — Direct thermal integration (non-separation): The steam-H_2 mixture is fed directly into a downstream **combustor** where air is injected to burn the hydrogen. The combustion further superheats the steam, which is then expanded through a turbine. This approach eliminates the need for hydrogen separation, purification, compression, and storage—all of which are expensive, energy-intensive, and hard to certify. This is the preferred pathway for a Generation-1 prototype.

Option B — Hydrogen separation (limited buffer only): The steam is condensed to liquid water, and the gaseous hydrogen is separated for later combustion. However, to minimise system complexity, weight, and certification risk, **hydrogen storage will be strictly limited to a small buffer**—sufficient for no more than 10 minutes of operation at full power. This buffer serves only to smooth transient mismatches between hydrogen production and demand, not as a primary energy store. **Metal hydride storage and large-scale compressed hydrogen storage are explicitly excluded** from consideration for the Generation-1 system, as they introduce prohibitive weight, cost, and certification burdens. The analysis will identify the minimum buffer size required to enable the selected prime mover, but will not design the storage system itself.

The trade-off analysis will compare the two options against the evaluation criteria defined in Step 1. The hypothesis—that Option A (direct thermal integration) is superior for a Generation-1 system—will be tested using the weighted scoring.

Step 3 — Prime Mover Candidate Selection

Based on the outcome of Step 2, a set of mature, commercially available prime movers compatible with the selected thermal integration pathway will be identified. For Option A (direct thermal integration), the candidate prime movers are:

1. **Steam turbines** — Mature technology with extensive marine and industrial service records. Well-matched to the high-temperature steam output. Class society rules are well established. Efficiency in the 20–30% range for small-scale units (100–500 kW).

2. **Organic Rankine Cycle (ORC) engines** — Use an organic working fluid with a lower boiling point than water, enabling efficient power extraction from lower-temperature heat sources. Commercially available in the 50–500 kW range (e.g., Turboden, Ormat, Enertime). Increasingly used in marine waste-heat recovery, but have less extensive marine certification history than steam turbines.

3. **Diesel engines (dual-fuel or hydrogen-enriched)** — Dominant prime mover in the existing marine fleet, with mature certification pathways. Integrating hydrogen into a diesel cycle introduces injection, combustion control, and emissions challenges (particularly NOx). This option will be retained primarily as a reference case and is expected to score lower than steam turbines on simplicity and certification risk.

Prime movers that are speculative, developmental, or lack marine certification will be **explicitly deprioritised**. These include solid oxide fuel cells, opposed-piston engines, supercritical CO_2 turbines, and Stirling engines. A brief justification for each deprioritised technology will be included in the thesis.

For each candidate prime mover, data on efficiency, power-to-weight, cost, maintenance intervals, and certification status will be gathered from:

- Manufacturer datasheets and technical publications.
- Classification society guidance documents (e.g., DNV-GL, LR, RINA class rules).
- Published literature on small-scale steam and ORC systems.
- Industry interviews (from Section 3.1).

Step 4 — Battery Buffer Specification

The transient response characterisation from Section 3.3 will produce a set of measured lag times, rise times, and ramp rates for the reactor when transitioning between power modes. These data will be used to define the requirements for a buffer battery.

Calculating the required energy and power:

For each measured mode transition, the battery's required energy and power are determined by the **difference** between what the vessel demands and what the reactor actually produces during the transient. The reactor is never producing zero power—it is always ramping from one level to another—so the battery only needs to supply the **deficit**, not the full demand.

For example, if the vessel demands 150 kW but the reactor is currently producing 50 kW and ramping up, the battery must supply the 100 kW difference. If the reactor takes 30 seconds to complete the ramp, the total energy the battery must deliver is the sum of that 100 kW deficit over the ramp period—significantly less than the energy that would be required if the reactor were completely off. The same logic applies in reverse: if the reactor overshoots and produces more power than demanded, the battery may absorb the excess (charging), which affects its cycle life and cooling requirements.

For each transition, the **peak power deficit** (the largest instantaneous gap between demand and reactor output) defines the required battery power rating (kW). The **total energy deficit** (the sum of the gap over the entire transient) defines the required battery energy capacity (kWh).

Identifying the worst-case transition:

The transition that yields the largest energy requirement defines the minimum energy capacity, while the transition that yields the largest instantaneous power deficit defines the minimum power rating. These worst-case transitions are not necessarily the same—a slow ramp over a long period may require more energy, while a sudden demand step with a long lag may require high instantaneous power.

Applying a safety margin:

A safety margin (e.g., 20%) will be applied to both the energy and power requirements to account for measurement uncertainty, degradation over the battery's lifetime, and conservative assumptions in the demand profiles.

Surveying commercially available marine batteries:

A systematic survey of marine-certified battery systems will be conducted. Focus will be on **lithium iron phosphate (LFP)** and **nickel manganese cobalt (NMC)** chemistries, as these have existing marine approvals (e.g., DNV-GL type approval, LR certification). Speculative chemistries (solid-state, sodium-ion, lithium-sulphur) announced in press releases but not yet commercially available in marine form will be excluded.

Recommending a hybrid topology:

The simplest hybrid architecture—series, parallel, or combined—that meets the buffering requirements will be recommended. The **series hybrid** (reactor to generator to battery to electric motor) decouples the reactor from the propeller, allowing the reactor to operate at its most efficient steady state while the battery handles transients. This is mechanically simple and will be the default recommendation unless a compelling case for parallel hybrid emerges.

Producing a procurement specification:

The final output will be a specification sheet listing the required battery capacity (kWh), power (kW), nominal voltage, C-rate, cooling requirements, and preferred chemistry. This specification is intended to be handed to a battery supplier for quotation, not to be a detailed engineering design.

Step 5 — Efficiency Target Relaxation

The preliminary energy density analysis in Section 1.1 assumed a 40% engine efficiency for metal fuels, derived from theoretical projections rather than real-world demonstration. For a Generation-1 prototype, this assumption is unrealistic. The methodology for revisiting the efficiency target is:

- **Literature review:** Gather published data on the actual efficiencies of small-scale (less than 1 MW) steam turbines, ORC engines, and diesel gensets. These will typically be in the range of 20–30% for steam turbines and ORC.
- **Consultation with manufacturers:** Where possible, obtain indicative efficiency curves from suppliers for units in the 100–500 kW range.
- **Trade-off analysis:** Compare the effective energy density of the aluminium fuel system (from Tables 1 and 2) using the literature-based efficiencies (20–30%) against competing electro-fuels (hydrogen, ammonia, methanol). If aluminium remains competitive (within a factor of 2 in volumetric energy density), the relaxed target is acceptable.
- **Justification:** The thesis will explicitly argue that a 20–30% efficiency target is sufficient for a proof-of-concept prototype, and that efficiency optimisation can be pursued in later development iterations once the fundamental safety and controllability of the reactor are demonstrated.

Step 6 — Synthesis into an Architectural Recommendation

The outputs of Steps 1–5 will be synthesised into a single, coherent **architectural recommendation** for the Generation-1 propulsion system. This recommendation will specify:

- The selected thermal integration pathway (direct combustion vs. limited separation).
- The recommended prime mover and the reason for its selection.
- The critical interface requirements: steam temperature, pressure, mass flow rate, hydrogen fraction, and allowable transient ramp rates.
- The recommended battery chemistry, capacity (kWh), and power (kW), and the preferred hybrid topology.
- The revised efficiency target and its justification.

The recommendation will be presented as a **decision matrix** and a **narrative summary**, with all assumptions and uncertainties explicitly documented. This is **not** a final engineering design—it is a **feasible and de-risked pathway** for the next phase of research and development.

Limitations and Scope Boundary

This analysis is explicitly **desk-based**. It does not involve:

- Simulation of engine or battery performance.
- Physical testing of any prime mover or battery cell.
- Detailed design of engine control systems, alternators, power electronics, or thermal management.
- Cost-engineering or procurement of any component.

The outputs are **requirements and recommendations**, not designs. They are intended to inform and de-risk subsequent work, which will involve full-scale simulation, prototype integration, and certification testing.

4 Proposed Project Time-line

The research timeline is a 24-month, risk-buffered plan built around four structural features:

- **Four go/no-go gates** — explicit decision points at which scope is adjusted if time has been lost.
- **A tiered experiment programme** — Tier 1 (core), Tier 2 (extended), and Tier 3 (optional). The thesis is defensible on Tier 1 alone.
- **Parallel development of control systems and engine/battery analysis** — control system development begins in parallel with later experiments, while the engine and battery trade-off study runs after the core experiments complete.

Prioritisation of Experimental Tiers

The experimental programme has been structured to prioritise the most critical data for the project's primary objectives:

- **Tier 1 (Core)** — Distilled water experiments covering temperature, pressure, particle size, and fuel-to-water ratio optimisation. This tier is essential for the project's success and provides all data required for control system development and battery sizing.
- **Tier 2 (Extended)** — Synthetic seawater experiments and limited natural seawater tests. These are valuable for understanding seawater's effects but are not required for the primary research objectives. Tier 2 is pursued if the experimental schedule permits.
- **Tier 3 (Optional)** — Extended natural seawater runs (both coarse-filtered and fine-filtered), long-duration fouling studies, and repeat validation runs. These are conducted only if the project is ahead of schedule.

The tiered structure ensures that the thesis remains defensible on Tier 1 alone, while providing flexibility to extend the work if time and resources permit.

Go/No-Go Gates

Gate 1 — Month 4: Safety Approval. No high-pressure or high-temperature work begins before university laboratory safety approval is granted. If approval is delayed by more than two weeks, particle size grades are reduced from seven to four to protect the experiment schedule.

Gate 2 — Month 7: Requirements Matrix. Power output targets and energy density thresholds must be confirmed before the main parameter sweep begins. If stakeholder interviews are delayed, a provisional matrix will be constructed from publicly available superyacht specifications and class rule documentation from RINA, Lloyd's Register, and DNV, and submitted to the supervisor for formal approval. Interviews continue in parallel and any refined targets are incorporated before Tier 2 experiments begin.

Gate 3 — Month 12: Core Experiments Complete. All Tier 1 experiments must be finished by this point. If not, the contingency buffer (built into Months 14–15) is used to complete them, and Tier 2 and 3 experiments are formally deferred to future work.

Gate 4 — Month 20: Control Validation and Engine Analysis Complete. The control system validation and the engine/battery trade-off study must be completed by this point. If either is delayed, the thesis writing phase (Months 21–24) remains available for completing analysis and writing, but the scope of the control or engine sections may need to be reduced.

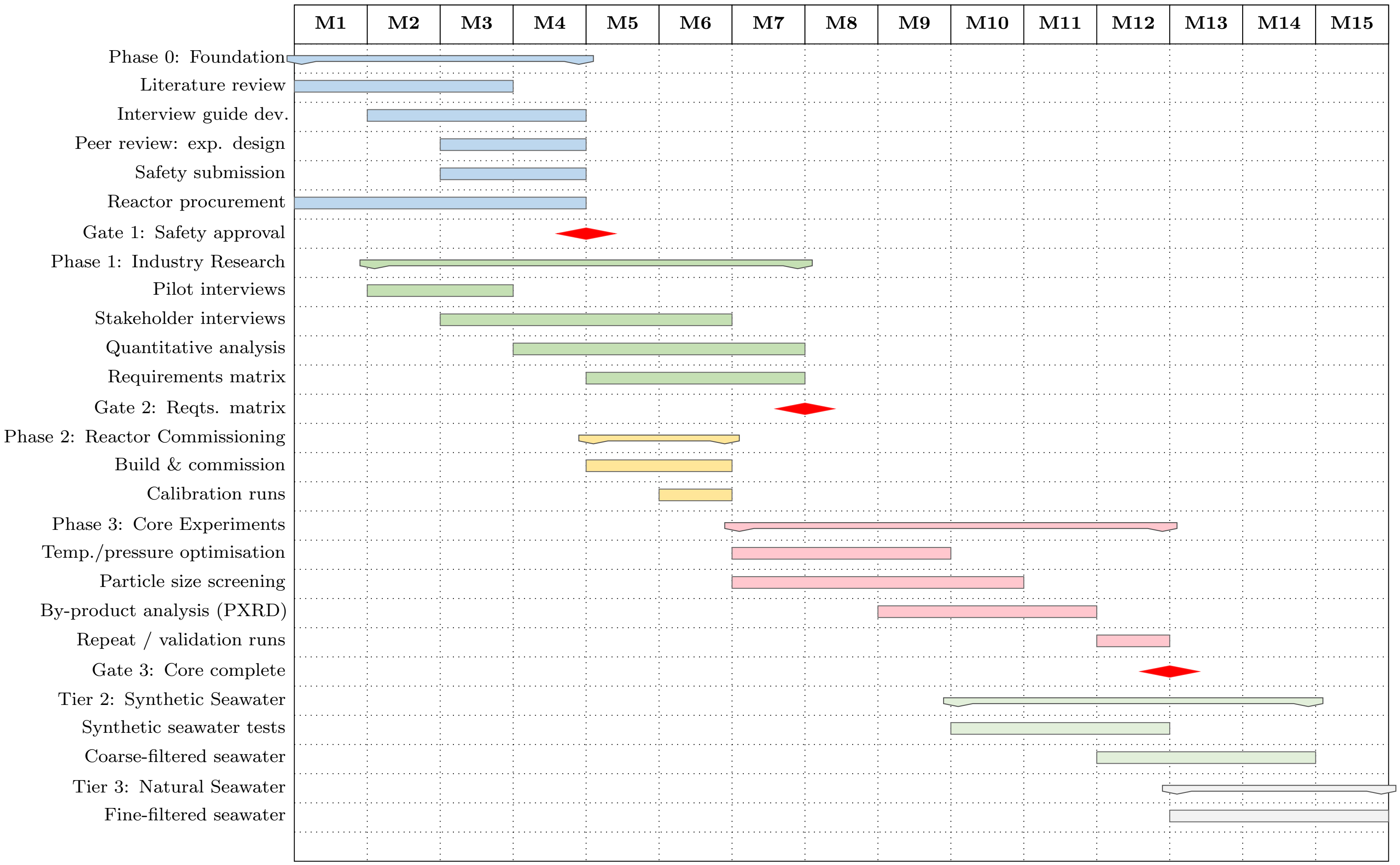
Key Scheduling Decisions

- **Procurement begins Month 2**, before safety approval. The pressure vessel and plunger pump carry 10–16 week lead times; waiting until Month 4 would push commissioning to Month 8 and collapse the experiment window.
- **Thesis writing is concentrated in Months 21–24**, after all experimental work and analysis is complete. However, literature review and preliminary chapter structuring begin earlier.
- **A 4-week contingency buffer** is built into Month 12 for repeat runs, instrument troubleshooting, and analysis backlogs.

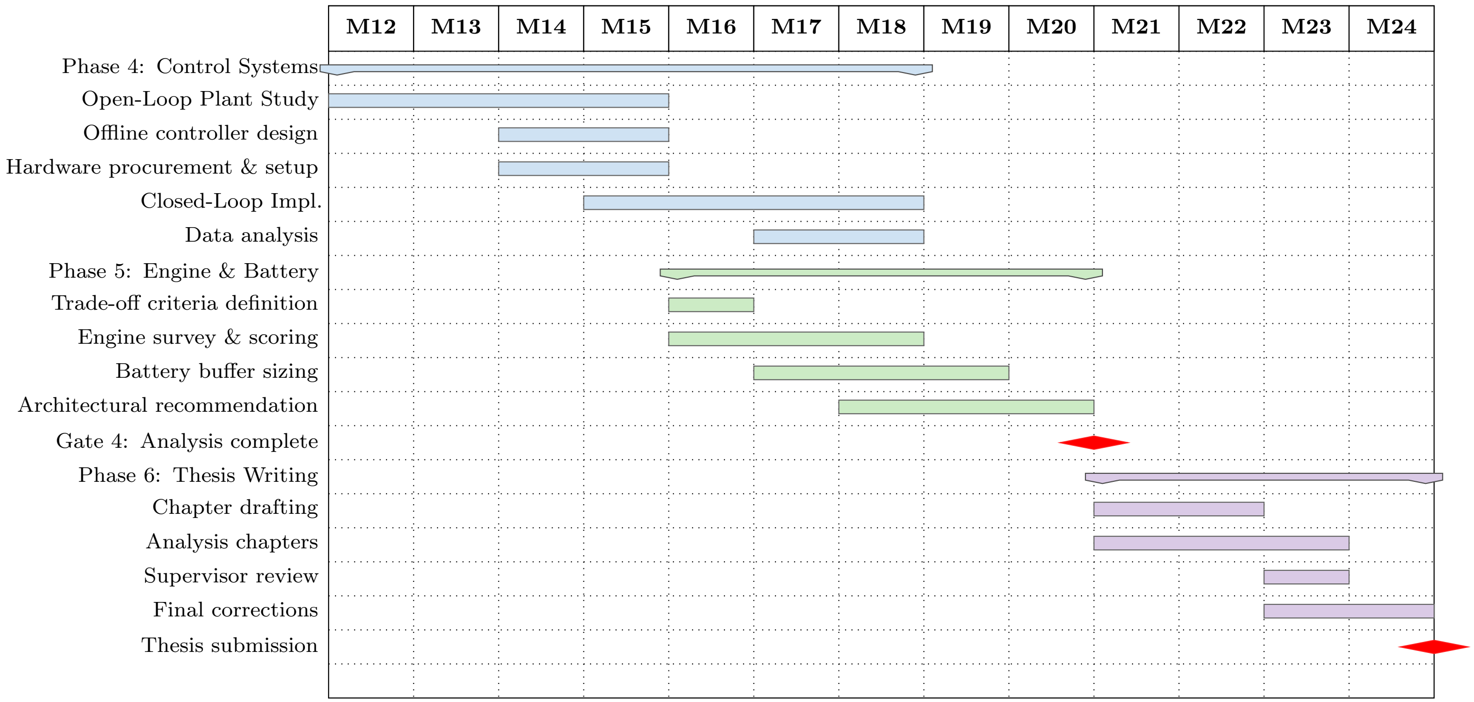
Go/No-Go Gate Summary

Gate	Month	Criterion	If not met
Gate 1	M4	Safety approval granted	Reduce particle size grades: $7 \to 4$
Gate 2	M7	Requirements matrix finalised	Use provisional targets; revise in thesis
Gate 3	M12	All core experiments complete	Tier 2/3 experiments become future work
Gate 4	M20	Control validation and engine/battery analysis complete	Reduce scope of control or engine sections
Submission	M24	Thesis submitted	—

Go/No-Go Gate Summary



M1
M2
M3
M4
M5
M6
M7
M8
M9
M10
M11
M12
M13
M14
M15
Phase 0: Foundation
Literature review
Interview guide dev.
Peer review: exp. design
Safety submission
Reactor procurement
Gate 1: Safety approval
Phase 1: Industry Research
Pilot interviews
Stakeholder interviews
Quantitative analysis
Requirements matrix
Gate 2: Reqts. matrix
Phase 2: Reactor Commissioning
Build & commission
Calibration runs
Phase 3: Core Experiments
Temp./pressure optimisation
Particle size screening
By-product analysis (PXRD)
Repeat / validation runs
Gate 3: Core complete
Tier 2: Synthetic Seawater
Synthetic seawater tests
Coarse-filtered seawater
Tier 3: Natural Seawater
Fine-filtered seawater


M12
M13
M14
M15
M16
M17
M18
M19
M20
M21
M22
M23
M24
Phase 4: Control Systems
Open-Loop Plant Study
Offline controller design
Hardware procurement & setup
Closed-Loop Impl.
Data analysis
Phase 5: Engine & Battery
Trade-off criteria definition
Engine survey & scoring
Battery buffer sizing
Architectural recommendation
Gate 4: Analysis complete
Phase 6: Thesis Writing
Chapter drafting
Analysis chapters
Supervisor review
Final corrections
Thesis submission

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Appendix A: Laboratory Safety Protocols and Experimental Plan Verification

A.1 Safety Protocols

The experimental programme involves four concurrent categories of hazard. Each must be fully addressed before any laboratory work begins.

A.1.1 Aluminium Powder Hazards

Micron-sized aluminium powder presents four distinct hazards:

1. **Dust cloud combustion.** Aluminium dust clouds are highly combustible and can detonate if ignited above the minimum ignition energy threshold.
    - All powder transfers will be conducted inside a dedicated enclosed glove box maintained under a dry nitrogen atmosphere.
    - Open flames, electrical sparks, and non-antistatic equipment are strictly prohibited within five metres of all powder handling areas.
2. **Exothermic reaction with moisture.** Aluminium reacts exothermically with ambient moisture, liberating flammable hydrogen gas.
    - All powder handling will be conducted under inert atmosphere or with active humidity control maintained below 30% relative humidity.
    - The reactor will be nitrogen-purged before each loading operation.
3. **Inhalation risk (aluminosis).** Chronic inhalation of aluminium dust causes pulmonary fibrosis.
    - P100 full-face respirators are mandatory during all powder handling.
    - An enclosed handling system is preferred wherever practicable.
    - Post-experiment reactor cleaning will be conducted under wet-suppression conditions to prevent dust re-entrainment.
4. **Prohibition on water-based extinguishers.** Water reacts with aluminium to produce hydrogen and is strictly prohibited as an extinguishing agent in any area where aluminium powder is present.
    - Class D dry-powder extinguishers (copper powder or sodium chloride type) will be stationed at all powder handling positions.

A.1.2 Hydrogen Gas Hazards

Hydrogen has a lower explosive limit (LEL) of 4% in air. The following controls will be maintained throughout all experimental sessions.

Ventilation and monitoring:

- A dedicated forced-ventilation exhaust hood will be installed over the reactor gas outlet, maintaining hydrogen concentration below 1% (25% of LEL) at all monitoring points.
- Electrochemical hydrogen sensors will be positioned in the gas outlet stream and at multiple points within the reactor enclosure.
- Audible alarms will activate at 1% hydrogen concentration.
- Automatic equipment shutdown interlocks will engage at 2%.

Ignition prevention:

- All equipment will be bonded and earthed to prevent static discharge ignition.
- Antistatic flooring will be installed throughout the reactor area.
- Synthetic clothing is prohibited during all experimental sessions.

Containment and disposal:

- All gas line connections will use pressure-rated compression fittings rated to at least twice operating pressure.
- A nitrogen leak check will be performed before every experiment.
- All hydrogen not captured for measurement will be routed through a catalytic oxidiser before venting to atmosphere, eliminating both the ignition risk and the indirect climate impact of hydrogen release.

A.1.3 High-Pressure and High-Temperature Hazards

Pressure vessel integrity:

- All pressurised components will be rated to a minimum of twice the maximum operating pressure ($\geq$50 MPa).
- The reactor vessel will undergo third-party hydrostatic pressure testing annually.
- A spring-loaded pressure relief valve set at 1.25× operating pressure will be installed, with an independent secondary burst disc as a redundant protection layer.

Operational controls:

- The reactor will be operated from behind a rated polycarbonate blast shield at all times.
- All reactor external surfaces will be thermally insulated.
- Heat-resistant gloves and a face shield are mandatory during setup and any access to the reactor area during operation.
- A formal cooldown and pressure equalisation protocol will be followed before any reactor valve is opened.
- An interlock will prevent opening any reactor valve while internal pressure exceeds 0.5 MPa.

Steam outlet management:

- All steam outlet lines will be thermally insulated and labelled with appropriate warning markings.
- All steam outlet lines will be routed through a condensate trap before reaching any gas measurement instruments.

A.1.4 Reactor Fouling Control and Cleaning Protocol

Background and justification. The experimental programme uses a continuous-flow architecture: aluminium-water slurry is injected continuously via the variable-speed high-pressure plunger pump, the aluminium-water reaction proceeds within the pressurised reactor vessel, and the product stream — a mixture of hydrogen gas, steam, and aluminium oxide slurry — exits continuously.

Al_2O_3 (corundum) has a Mohs hardness of approximately 9, making it one of the most abrasive ceramic materials encountered in process engineering [62]. In a continuous-flow system it forms continuously at the reaction interface and must be carried out of the reactor by the product stream. The critical fouling locations are:

- **Product outlet valve and withdrawal line** — progressive blockage reduces product withdrawal flow rate and causes reactor pressure to rise at constant injection rate.
- **Slurry injection nozzle** — erosive wear progressively enlarges the nozzle bore, changing the injection flow rate at any given pump speed.
- **Plunger pump head and check valves** — continuous exposure to abrasive slurry at high pressure accelerates wear of valve seats and the plunger seal.

If fouling is not actively managed, it introduces drift in injection rate and reactor pressure during experimental runs, constituting an uncontrolled variable that directly corrupts transient load data [63].

Pre-experiment fouling baseline characterisation

Before the main parameter sweep begins, three to five continuous-flow baseline runs will be conducted at representative steady-state conditions:

- Temperature: 350°C Pressure: 20 MPa
- Particle size: 100 µm Water-to-aluminium mass ratio: 9
- Run duration: minimum 60 minutes at constant pump speed

The following measurements will be recorded at 5-minute intervals throughout each baseline run:

Parameter	Instrument	Fouling indicator
Pump delivery flow rate	Pump speed vs. calibration curve	Injection nozzle or check valve blockage
Reactor pressure	High-pressure transducer	Product outlet blockage
Gas outlet flow rate	Thermal mass flow controller	Product withdrawal line restriction

The rate of drift in each parameter per hour of continuous operation establishes:

1. The maximum safe run duration before a flush is required.
2. Which fouling location is the binding constraint under these conditions.

This is determined empirically before the parameter sweep begins, so that flush intervals are set by evidence rather than assumption.

Between-condition flush protocol

A flush will be performed at the end of every experimental condition — defined as a complete steady-state run plus any transient load modulation sequence at a given combination of temperature, pressure, particle size, and fuel-to-water ratio — before changing to the next set of conditions.

Step 1. Water purge (pressurised). Cease aluminium slurry injection. Continue pumping deionised water at the same flow rate for a minimum of 10 minutes to flush residual Al_2O_3 slurry from the reactor body, product outlet line, and withdrawal valve.

Step 2. Controlled cool-down. Reduce reactor temperature to below 200°C using a controlled heater ramp-down, while maintaining pressure via the back-pressure regulator to prevent flash vaporisation of residual water in the product line.

Step 3. Controlled depressurisation. Once below 200°C, slowly equalise reactor pressure to atmospheric through the back-pressure regulator over a minimum of 15 minutes.

Step 4. Alkaline flush. Circulate a dilute alkaline flush solution (1–2% NaOH in deionised water) at low temperature (<60°C) and atmospheric pressure through the reactor body, product outlet line, and withdrawal valve for 30 minutes. This exploits the amphoteric chemistry of aluminium oxide:

$$Al_2O_3 + 2\,NaOH \longrightarrow 2\,NaAlO_2 + H_2O \tag{3}$$

dissolving deposited Al_2O_3 and $Al(OH)_3$ from internal surfaces [64]. The NaOH concentration is deliberately conservative to avoid corrosive attack on the SS316L reactor body.

Step 5. Deionised water rinse. Flush with a minimum of five system volumes of deionised water. The pH of the final rinse water must be within ± 0.2 of the deionised water baseline (pH $\approx$ 6.5–7.0) before re-pressurisation begins.

Step 6. Blank verification run. Conduct a continuous-flow run using deionised water only (no aluminium) at the target temperature and pressure for the next experimental condition, to:

- Confirm complete removal of alkaline residue.
- Re-establish the thermal steady-state baseline before slurry injection resumes.

Waste Disposal Note

The spent $NaOH/NaAlO_2$ flush solution is a dilute alkaline waste stream. It will be collected in a labelled chemical waste container and disposed of in accordance with the university's chemical waste management procedures. Chemical-resistant gloves and eye protection are mandatory during all flushing operations.

Pump head and injection nozzle inspection

Component	Frequency	Threshold	Action
Pump delivery flow rate	Start of every session	$> \pm 3\%$ from calibration	Pump head disassembly and inspection
Check valves and plunger seal	End of every experimental day	Any visible wear or deposit	Component replacement
Injection nozzle bore diameter	Start of every experimental week	$> 5\%$ increase from original spec	Nozzle replacement

Note: The pump flow rate tolerance of ±3% is tighter than used in intermittent operation (typically ±5%) because the transient load characterisation experiments depend on accurate, stable injection rates throughout sustained continuous-flow runs [63].

Sensor drift monitoring during continuous-flow runs

Because transient load data is collected in real time during sustained runs, sensor drift must be detected and bounded *during* a run, not only between runs:

1. **Pressure transducers.** The pump inlet transducer and reactor pressure transducer will be cross-checked against each other at the start and end of every run.
   - Divergence of $> \pm 2\%$ of full scale at steady state → flush before next run.
2. **Hydrogen yield cross-validation.** The mass flow controller reading on the gas outlet will be cross-validated against the pressure-time derivative method ($\mathrm{d}P/\mathrm{d}t$) at steady state at the start of each run.
   - Discrepancy of $> \pm 5\%$ between the two methods → terminate run, inspect system for fouling or developing leak before continuing.
3. **Thermocouple array.** The four axial thermocouples will be cross-checked against each other at steady state.
   - Any thermocouple deviating $> \pm 5$°C from the profile established during the blank calibration run at the same temperature setpoint → flag for inspection at end of that run.

A.1.5 General Laboratory Controls

- A minimum of two researchers must be present during all experimental sessions.
- A dedicated safety observer will be assigned for each session, responsible for monitoring all sensor readings in real time and activating the emergency shutdown procedure if any parameter exceeds its safe operating limit.

- Emergency shutdown procedures will be formally documented and prominently displayed within the laboratory.
- All researchers will complete formal training on emergency procedures before participating in any experimental work.

A.2 Experimental Plan Verification and University Approval

The experimental programme described in Section 3.2 represents the high-level research plan appropriate for a Master's proposal. Before any laboratory work begins, this plan must undergo two independent layers of review.

A.2.1 Internal Peer Review

Process:

1. Following completion of the industry research phase (Months 1–4), the detailed experimental design will be submitted for internal peer review. This submission will include:
   - Reactor vessel specification.
   - Instrumentation layout and calibration methodology.
   - Full experimental sequences for all Tier 1, 2, and 3 experiment streams.
   - The complete safety protocol from Section A.1, including the fouling control procedures in Section A.1.4.
2. Review will be conducted by a panel of at least two independent academics from the university's mechanical and chemical engineering departments with direct experience in high-pressure reactor systems.
3. All reviewer comments will be formally documented and the experimental plan revised accordingly before being finalised.

Purpose: To identify blind spots in experimental design, instrumentation, and safety provisions that may not be apparent from within the research team.

A.2.2 University Laboratory Safety Approval

Submission requirements:

A comprehensive laboratory experimental proposal will be submitted to the university's laboratory safety committee and the relevant departmental review body. This submission will include:

- Full safety protocol (this appendix, as revised).
- Risk register covering all four hazard categories in Section A.1.
- Emergency response and shutdown procedures.
- Chemical waste disposal plan for spent NaOH/$NaAlO_2$ flush solution and aluminium oxide solid waste.
- Detailed reactor specifications including pressure vessel ratings, relief valve sizing, and instrumentation layout.
- Evidence of compliance with applicable pressure vessel standards.
- Reference to the safety framework developed in consultation with RINA, Lloyd's Register, and DNV (Section 2.2).

Hard Stop

No high-pressure or high-temperature experimental work will commence before this approval is formally granted.

Target date: University approval will be completed by the end of Month 4 (Quarter 1 of the research programme), consistent with the Gate 1 milestone in the project timeline.

Appendix B: Energy Density Analysis Code

B.1 Python Analysis Script

```
#!/usr/bin/env python3
import json
import matplotlib.pyplot as plt  # pyright: ignore[reportMissingImports]
from adjustText import adjust_text  # pyright:
↪  ignore[reportMissingImports]

# Load the JSON file
with open('fuel-dataset.json', 'r') as f:
    data = json.load(f)

# Extract fuel data
fuels = data['fuels']

```

```python
# Calculate effective mass and volumetric energy densities
fuel_names = []
mass_energy_densities = []
volumetric_energy_densities = []

eff_mass_energy_densities = []
eff_volumetric_energy_densities = []

categories = []

for fuel_key, fuel_data in fuels.items():
    name = fuel_data['name']

    # Skip Coal, H2_Gas, and fuels with metal oxide in name
    if name == "Coal" or name == "H2_Gas" or name == "CNG":
        continue
    if "->" in name:  # Skip metal oxide combustion fuels (e.g., "MgH2->MgO")
        continue
    # Skip pure metal oxides (Fe3O4, AlOOH, Al2O3, MgO, B2O3, SiO2, ZnO)
    if "O" in name and any(char.isdigit() or char == "O" for char in name) and "_ref" not in name:
        # Check if it's a metal oxide (contains O and numbers/O, but not a reference)
        if name in ["Fe3O4", "AlOOH", "Al2O3", "MgO", "B2O3", "SiO2", "ZnO"]:
            continue

    density = fuel_data['density']  # kg/l
    mass_energy = fuel_data['mass_energy_density']  # kWh/kg
    efficiency = fuel_data['engine_efficiency']

    # Calculate volumetric energy density
    vol_energy = mass_energy * density

    # Clean up metal names
    display_name = name.replace(" (ref)", "")

    # Replace full metal names with symbols
    metal_symbols = {
        "Iron": "Fe",
```

```
        "Aluminium": "Al",
        "Magnesium": "Mg",
        "Boron": "B",
        "Si": "Si",
        "Zinc": "Zn"
    }
    for metal, symbol in metal_symbols.items():
        if display_name == metal:
            display_name = symbol
            break

    # Simplify other fuel names
    name_simplifications = {
        "Pressure_Tank_H2": "H2",
        "Liquid_NH3": "NH3",
        "Heavy Fuel Oil": "HFO"
    }
    if display_name in name_simplifications:
        display_name = name_simplifications[display_name]

    # Default calculation
    eff_mass_energy_density = mass_energy * efficiency
    eff_volumetric_energy_density = vol_energy * efficiency

    # Determine category for coloring
    if display_name in ["H2", "NH3"]:
        category = "Simple Electro-fuels"
    elif display_name in ["Diesel", "Petrol", "Methanol", "Ethanol",
    ↪  "LNG", "LPG"]:
        category = "Synthetic Hydrocarbons"
    elif display_name in ["Fe", "Al", "Mg", "B", "Si", "Zn"]:
        category = "Simple Metal Fuels"
    elif display_name in ["MgH2", "AlH3", "LiAlH4", "NaBH4", "LiBH4",
    ↪  "NaAlH4"]:
        category = "Metal Hydrides"
    elif display_name == "HFO":
        category = "HFO"
    else:
        category = "Other"

    fuel_names.append(display_name)
```

```
    mass_energy_densities.append(mass_energy)
    volumetric_energy_densities.append(vol_energy)
    eff_mass_energy_densities.append(eff_mass_energy_density)
    eff_volumetric_energy_densities.append(eff_volumetric_energy_density)
    categories.append(category)

# Create scatter plot
color_map = {
    "Simple Electro-fuels": "#3498db",  # Blue
    "Synthetic Hydrocarbons": "#e74c3c",   # Red
    "Simple Metal Fuels": "#2ecc71",    # Green
    "Metal Hydrides": "#f39c12", # Orange
    "HFO": "#9b59b6"             # Purple
}

plt.figure(figsize=(12, 8))
for category in color_map.keys():
    mask = [cat == category for cat in categories]
    x = [mass_energy_densities[i] for i in range(len(mask)) if mask[i]]
    y = [volumetric_energy_densities[i] for i in range(len(mask)) if mask[i]]
    plt.scatter(x, y, s=50, alpha=0.6, color=color_map[category], label=category)

plt.legend(loc='best', fontsize=10)

# Add labels for each point with adjustText to avoid overlap
# Calculate offset distance based on data range
x_range = max(mass_energy_densities) - min(mass_energy_densities)
y_range = max(volumetric_energy_densities) - min(volumetric_energy_densities)
offset_x = x_range * 0.00375  # 0.375% of x range
offset_y = y_range * 0.00375  # 0.375% of y range

texts = []
for i, name in enumerate(fuel_names):
    # Start text at an offset position from the point
    txt = plt.text(mass_energy_densities[i] + offset_x,
                   volumetric_energy_densities[i] + offset_y,
                   name,
                   fontsize=11, fontweight='bold', alpha=0.8)
```

```python
    texts.append(txt)

# Adjust text positions to avoid overlap with much larger spacing
adjust_text(texts,
            mass_energy_densities, volumetric_energy_densities,
            arrowprops=dict(arrowstyle='->', color='gray', lw=0.5, alpha=0.6),
            expand_points=(1.25, 1.25),
            expand_text=(1, 1),
            force_points=0.625,
            force_text=0.625,
            lim=50,
            only_move={'points':'xy', 'text':'xy'})

plt.xlabel('Mass Energy Density (kWh/kg)', fontsize=12)
plt.ylabel('Volumetric Energy Density (kWh/L)', fontsize=12)
plt.title('Fuel Energy Density Comparison', fontsize=14, fontweight='bold')
plt.grid(True, alpha=0.7, linestyle='-', linewidth=1.2, color='grey')
plt.savefig('fuel_energy_density.png', dpi=300, bbox_inches='tight')
plt.savefig('../images/simplified_fuel_energy_density.png', dpi=300, bbox_inches='tight')
print("\nPlot saved as 'fuel_energy_density.png'")

# Print summary statistics
print("\nFuel Energy Density Summary:")
print("=" * 70)
print(f"{'Fuel':<25} {'Mass (kWh/kg)':<15} {'Volumetric (kWh/L)':<20}")
print("=" * 70)
for i in range(len(fuel_names)):
    print(f"{fuel_names[i]:<25} {mass_energy_densities[i]:<15.2f} {volumetric_energy_densities[i]:<20.2f}")
print("=" * 70)

# Adjust metal fuels to use their oxide effective energy densities
metal_oxide_map = {
    "Fe": "Fe3O4",
    "Al": ["Al2O3", "AlOOH"],  # Al has two oxides
    "Mg": "MgO",
    "B": "B2O3",
    "Si": "SiO2",
```

```
    "Zn": "ZnO"
}

# For Al, we'll need to process both oxides separately
adjusted_fuel_names = []
adjusted_eff_mass_energy = []
adjusted_eff_vol_energy = []
adjusted_categories = []

for i, name in enumerate(fuel_names):
    if name in metal_oxide_map:
        # This is a simple metal fuel, replace with oxide(s)
        oxide_names = metal_oxide_map[name]
        if not isinstance(oxide_names, list):
            oxide_names = [oxide_names]

        for oxide_name in oxide_names:
            # Find the oxide in the original fuels data
            oxide_fuel = next((fuels[k] for k in fuels if
            ↪ fuels[k]['name'] == oxide_name), None)
            if oxide_fuel:
                oxide_density = oxide_fuel['density']
                oxide_mass_energy = oxide_fuel['mass_energy_density']
                oxide_efficiency = oxide_fuel['engine_efficiency']

                adjusted_fuel_names.append(oxide_name)
                adjusted_eff_mass_energy.append(oxide_mass_energy *
                ↪ oxide_efficiency)
                adjusted_eff_vol_energy.append(oxide_mass_energy *
                ↪ oxide_efficiency * oxide_density)
                adjusted_categories.append(categories[i])
    else:
        # Keep non-metal fuels as is
        adjusted_fuel_names.append(name)
        adjusted_eff_mass_energy.append(eff_mass_energy_densities[i])

        ↪ adjusted_eff_vol_energy.append(eff_volumetric_energy_densities[i])
        adjusted_categories.append(categories[i])

# Replace the original lists with adjusted ones for the second plot
fuel_names_for_plot2 = adjusted_fuel_names
```

```
eff_mass_energy_densities = adjusted_eff_mass_energy
eff_volumetric_energy_densities = adjusted_eff_vol_energy
categories_for_plot2 = adjusted_categories

plt.figure(figsize=(12, 8))
for category in color_map.keys():
    mask = [cat == category for cat in categories_for_plot2]
    x = [eff_mass_energy_densities[i] for i in range(len(mask)) if mask[i]]
    y = [eff_volumetric_energy_densities[i] for i in range(len(mask)) if mask[i]]
    plt.scatter(x, y, s=50, alpha=0.6, color=color_map[category], label=category)

plt.legend(loc='best', fontsize=10)

# Add labels for each point with adjustText to avoid overlap
# Calculate offset distance based on data range
x_range = max(eff_mass_energy_densities) - min(eff_mass_energy_densities)
y_range = max(eff_volumetric_energy_densities) - min(eff_volumetric_energy_densities)
offset_x = x_range * 0.00375  # 0.375% of x range
offset_y = y_range * 0.00375  # 0.375% of y range

texts = []
for i, name in enumerate(fuel_names_for_plot2):
    # Start text at an offset position from the point
    txt = plt.text(eff_mass_energy_densities[i] + offset_x,
                   eff_volumetric_energy_densities[i] + offset_y,
                   name,
                   fontsize=11, fontweight='bold', alpha=0.8)
    texts.append(txt)

# Adjust text positions to avoid overlap with much larger spacing
adjust_text(texts,
            eff_mass_energy_densities, eff_volumetric_energy_densities,
            arrowprops=dict(arrowstyle='->', color='gray', lw=0.5, alpha=0.6),
            expand_points=(1.25, 1.25),
            expand_text=(1, 1),
            force_points=0.625,
```

```
            force_text=0.625,
            lim=50,
            only_move={'points':'xy', 'text':'xy'})

plt.xlabel('Eff. Mass Energy Density (kWh/kg)', fontsize=12)
plt.ylabel('Eff. Volumetric Energy Density (kWh/L)', fontsize=12)
plt.title('Eff. Fuel Energy Density Comparison', fontsize=14, fontweight='bold')
plt.grid(True, alpha=0.7, linestyle='-', linewidth=1.2, color='grey')
plt.savefig('eff_fuel_energy_density.png', dpi=300, bbox_inches='tight')
plt.savefig('../images/eff_fuel_energy_density.png', dpi=300, bbox_inches='tight')
print("\nPlot saved as 'eff_fuel_energy_density.png'")
```

B.2 Fuel Dataset

```
{
  "fuels": {
    "HFO": {
      "name": "Heavy Fuel Oil",
      "density": 0.98,
      "mass_energy_density": 11.6,
      "engine_efficiency": 0.5,
      "source": "https://www.engineeringtoolbox.com/co2-emission-fuels-d_1085.html"
    },
    "Coal": {
      "name": "Coal",
      "density": 1.3,
      "mass_energy_density": 6.67,
      "engine_efficiency": 0.25,
      "source": "https://www.engineeringtoolbox.com/co2-emission-fuels-d_1085.html",
      "notes": "Density calculated as average of 1100-1500 kg/m³"
    },
    "H2_Gas": {
      "name": "H2_Gas",
      "density": 0.00009,
      "mass_energy_density": 39.44,
      "engine_efficiency": 0.5,
```

```json
      "notes": "Gaseous hydrogen at standard conditions"
    },
    "Pressure_Tank_H2": {
      "name": "Pressure_Tank_H2",
      "density": 0.027,
      "mass_energy_density": 1.58,
      "engine_efficiency": 0.5,
      "notes": "Hydrogen in pressure tank, effective energy density is 4% of H2 energy density"
    },
    "NH3": {
      "name": "Liquid_NH3",
      "density": 0.696,
      "mass_energy_density": 6.94,
      "engine_efficiency": 0.5,
      "notes": "Liquid ammonia, energy density is 17.6% of H2 energy density"
    },
    "CNG": {
      "name": "CNG",
      "density": 0.16,
      "mass_energy_density": 14.89,
      "engine_efficiency": 0.5
    },
    "LNG": {
      "name": "LNG",
      "density": 0.5,
      "mass_energy_density": 14.89,
      "engine_efficiency": 0.5
    },
    "Methanol": {
      "name": "Methanol",
      "density": 0.791,
      "mass_energy_density": 5.5,
      "engine_efficiency": 0.5
    },
    "LPG": {
      "name": "LPG",
      "density": 0.537,
      "mass_energy_density": 13.7,
      "engine_efficiency": 0.5
```

```
    },
    "Fe_ref": {
      "name": "Iron (ref)",
      "density": 7.9,
      "mass_energy_density": 2.05,
      "engine_efficiency": 0.4,
      "notes": "Iron fuel with no by-products"
    },
    "Fe3O4": {
      "name": "Fe3O4",
      "density": 5.15,
      "mass_energy_density": 1.49,
      "engine_efficiency": 0.4,
      "notes": "Iron fuel with Fe3O4 as by-product, energy density is Fe_ref / 1.38"
    },
    "Al_ref": {
      "name": "Aluminium (ref)",
      "density": 2.7,
      "mass_energy_density": 8.63,
      "engine_efficiency": 0.4,
      "notes": "Aluminium fuel with no by-products"
    },
    "AlOOH": {
      "name": "AlOOH",
      "density": 3.03,
      "mass_energy_density": 3.89,
      "engine_efficiency": 0.4,
      "notes": "Aluminium fuel with AlOOH as by-product, energy density is Al_ref / 2.22"
    },
    "Al2O3": {
      "name": "Al2O3",
      "density": 3.97,
      "mass_energy_density": 4.57,
      "engine_efficiency": 0.4,
      "notes": "Aluminium fuel with Al2O3 as by-product, energy density is Al_ref / 1.89"
    },
    "Mg_ref": {
      "name": "Magnesium (ref)",
```

```
      "density": 1.7,
      "mass_energy_density": 6.88,
      "engine_efficiency": 0.4,
      "notes": "Magnesium fuel with no by-products"
    },
    "MgO": {
      "name": "MgO",
      "density": 3.58,
      "mass_energy_density": 4.12,
      "engine_efficiency": 0.4,
      "notes": "Magnesium fuel with MgO as by-product, energy density is
       ↪  Mg_ref / 1.67"
    },
    "B_ref": {
      "name": "Boron (ref)",
      "density": 2.48,
      "mass_energy_density": 17.702,
      "engine_efficiency": 0.4,
      "notes": "Boron fuel with no by-products"
    },
    "B2O3": {
      "name": "B2O3",
      "density": 2.56,
      "mass_energy_density": 5.56,
      "engine_efficiency": 0.4,
      "notes": "Boron fuel with B2O3 as by-product, energy density is
       ↪  B_ref / 3.1818"
    },
    "Si_ref": {
      "name": "Si (ref)",
      "density": 2.33,
      "mass_energy_density": 9.01,
      "engine_efficiency": 0.4,
      "notes": "Silicon fuel with no by-products",
      "source": "doi:10.1016/j.energy.2005.12.001"
    },
    "SiO2": {
      "name": "SiO2",
      "density": 2.65,
      "mass_energy_density": 4.21,
      "engine_efficiency": 0.4,
```

```
      "notes": "Silicon fuel with SiO2 as by-product, energy density is Si_ref / 2.14"
    },
    "MgH2": {
      "name": "MgH2",
      "density": 1.46,
      "mass_energy_density": 3.0,
      "engine_efficiency": 0.5,
      "notes": "Metal hydride, energy density is 7.6% of H2_Gas",
      "source": "https://next-gen.materialsproject.org/materials/mp-23710?formula=MgH2&has_props=dos"
    },
    "AlH3": {
      "name": "AlH3",
      "density": 1.52,
      "mass_energy_density": 3.98,
      "engine_efficiency": 0.5,
      "notes": "Metal hydride, energy density is 10.1% of H2_Gas",
      "source": "10.17188/1199814"
    },
    "LiBH4": {
      "name": "LiBH4",
      "density": 0.72,
      "mass_energy_density": 7.3,
      "engine_efficiency": 0.5,
      "notes": "Metal hydride, energy density is 18.5% of H2_Gas",
      "source": "https://next-gen.materialsproject.org/materials/mp-30209?formula=LiBH4&has_props=dos"
    },
    "NaAlH4": {
      "name": "NaAlH4",
      "density": 1.34,
      "mass_energy_density": 2.96,
      "engine_efficiency": 0.5,
      "notes": "Metal hydride, energy density is 7.5% of H2_Gas",
      "source": "https://next-gen.materialsproject.org/materials/mp-23704?formula=NaAlH4&has_props=dos#properties"
    },
    "MgH2_MgO": {
      "name": "MgH2->MgO",
      "density": 3.58,
```

```
      "mass_energy_density": 6.0754,
      "engine_efficiency": 0.4,
      "notes": "Metal hydride combustion with MgO as by-product"
    },
    "AlH3_AlOOH": {
      "name": "AlH3->AlOOH",
      "density": 3.03,
      "mass_energy_density": 5.8737,
      "engine_efficiency": 0.4,
      "notes": "Metal hydride combustion with AlOOH as by-product"
    },
    "AlH3_Al2O3": {
      "name": "AlH3->Al2O3",
      "density": 3.97,
      "mass_energy_density": 6.9102,
      "engine_efficiency": 0.4,
      "notes": "Metal hydride combustion with Al2O3 as by-product"
    },
    "Zn_ref": {
      "name": "Zinc (ref)",
      "density": 7.14,
      "mass_energy_density": 1.48,
      "engine_efficiency": 0.4,
      "notes": "Zinc fuel with no by-products"
    },
    "ZnO": {
      "name": "ZnO",
      "density": 5.61,
      "mass_energy_density": 0.4748,
      "engine_efficiency": 0.4,
      "notes": "Zinc fuel with ZnO as by-product, energy density is Zn_ref / 1.2462"
    }
  },
  "metadata": {
    "description": "Fuel properties for shipping analysis",
    "units": {
      "density": "kg/l",
      "mass_energy_density": "kWh/kg",
      "engine_efficiency": "dimensionless (0-1)"
    },
```

```
    "notes": [
      "Effective mass energy density = mass_energy_density × engine_efficiency",
      "Effective volumetric energy density = effective_mass_energy_density × density",
      "Metal fuel energy densities from https://www.vibgyorpublishers.org/content/ijmmp/ijmmp-5-056-table2.html",
      "Metal densities from PubChem for individual chemicals"
    ]
  }
}
```